



(12) **United States Patent**
Loheide et al.

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(54) **PUBLISHING A DISPARATE PER-CLIENT LIVE MEDIA OUTPUT STREAM BASED ON DYNAMIC INSERTION OF TARGETED NON-PROGRAMMING CONTENT AND CUSTOMIZED PROGRAMMING CONTENT**

(58) **Field of Classification Search**
CPC H04N 21/235; H04N 21/2393; H04N 21/25841; H04N 21/26258; H04N 21/668;
(Continued)

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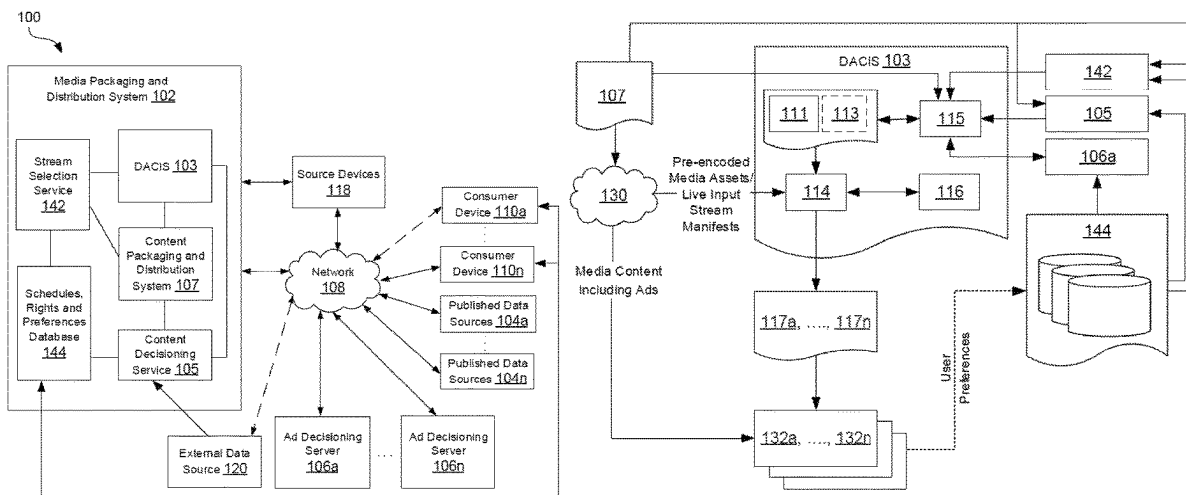
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(57) **ABSTRACT**
A system is provided for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content. The system generates a first programming schedule that defines locations and types of one or more decision points defined by one or more content placement opportunities in an indexed metadata based on one or more factors. The system further controls publishing of one or more pre-encoded media asset manifests as a first disparate live media output stream manifest based on a schedule of one or more pre-encoded media assets and the one or more decision points defined in the first programming schedule.

Related U.S. Application Data

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(51) **Int. Cl.**
H04N 21/234 (2011.01)
H04N 21/2187 (2011.01)
(Continued)
(52) **U.S. Cl.**
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39 Claims, 9 Drawing Sheets



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(58) **Field of Classification Search**

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See application file for complete search history.

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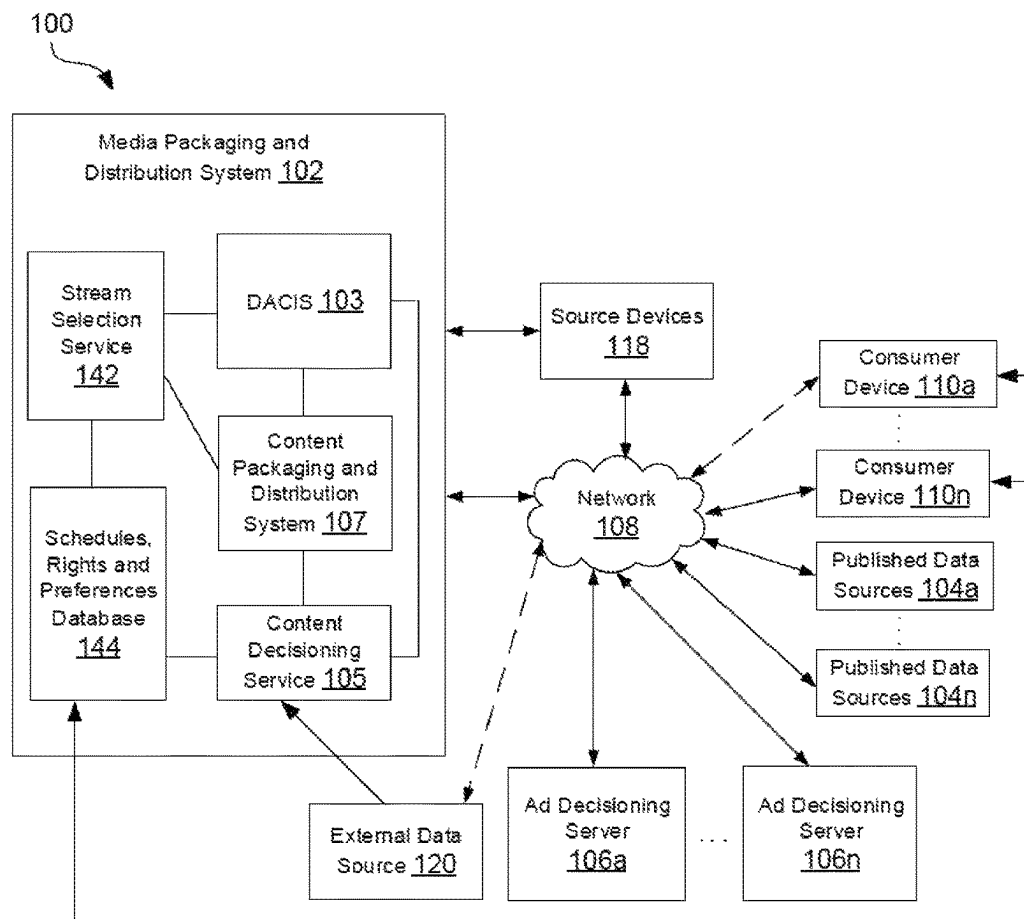


FIG. 1A

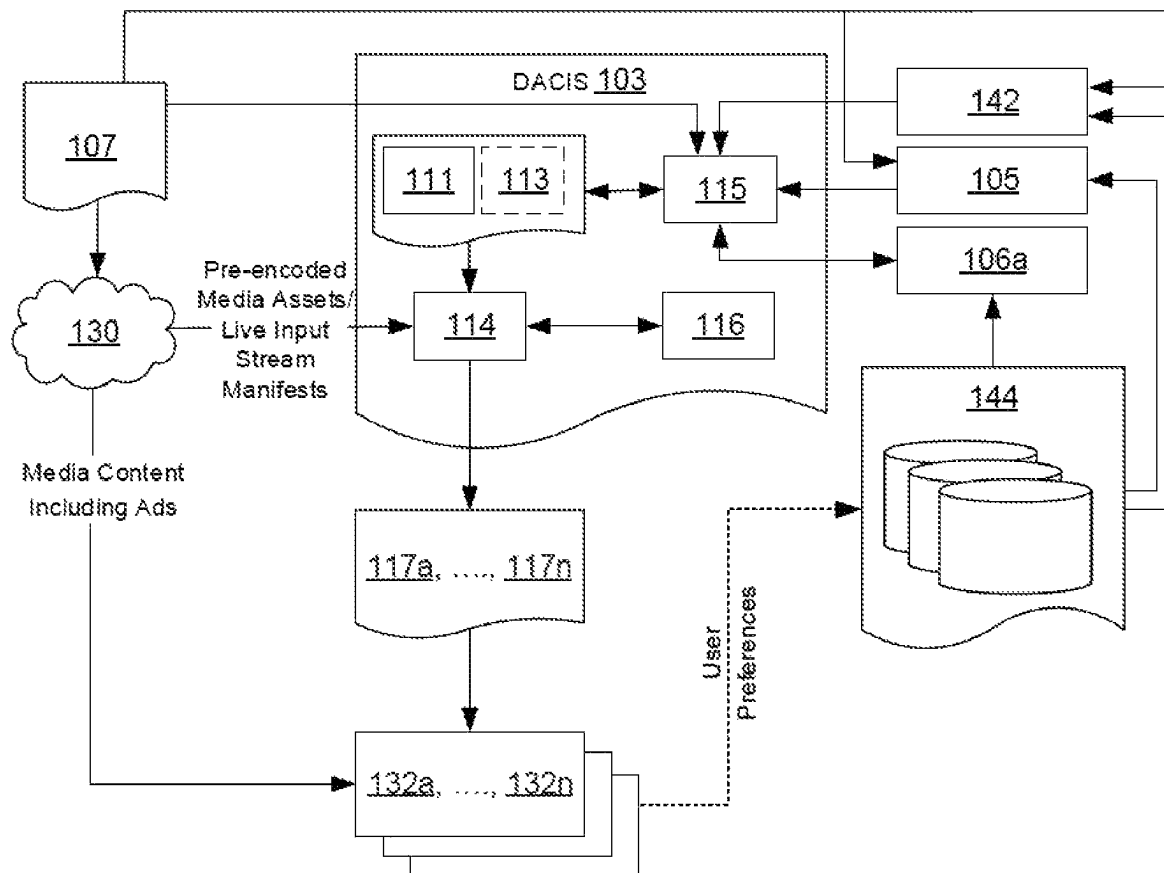


FIG. 1B

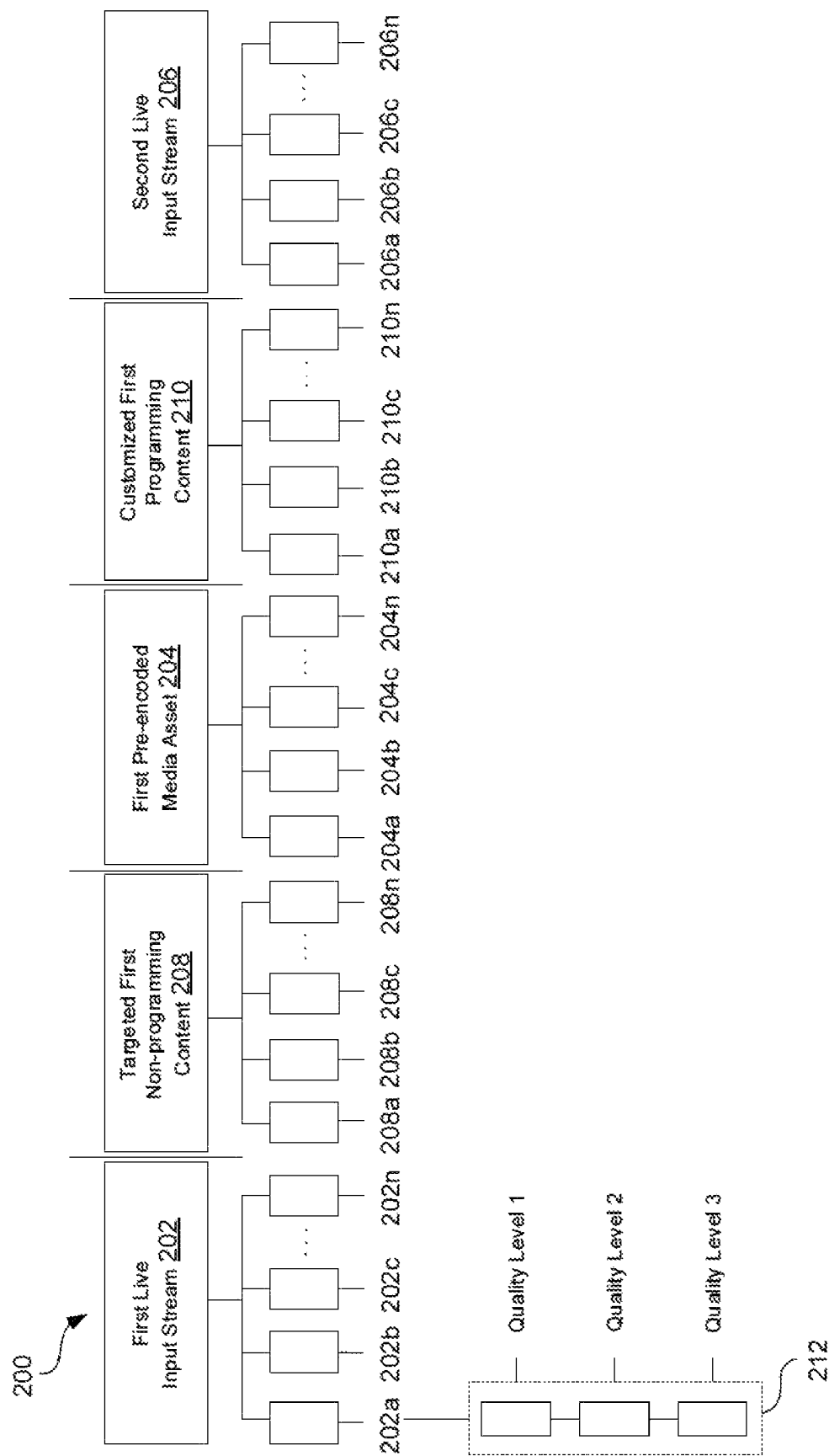


FIG. 2

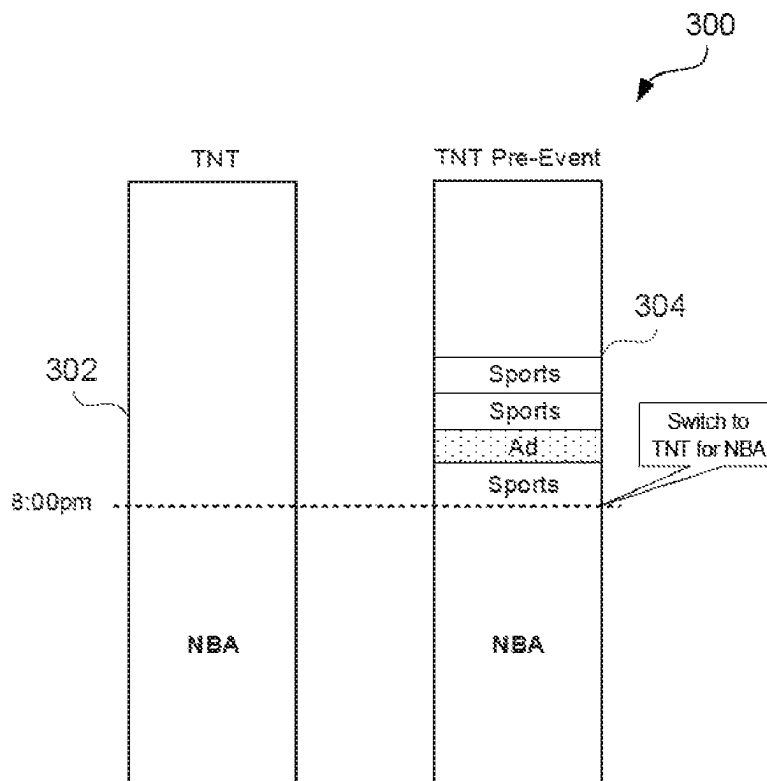


FIG. 3

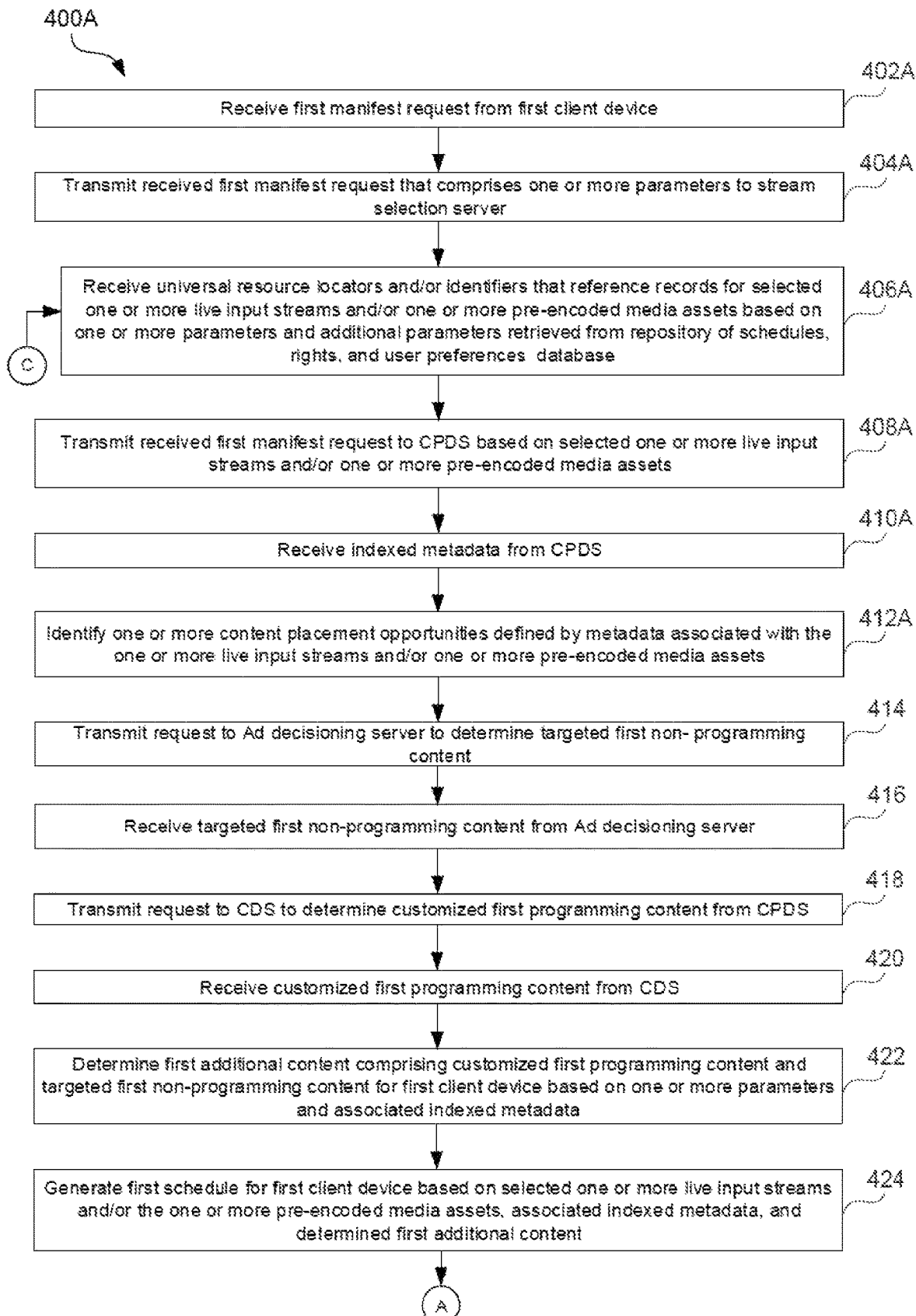


FIG. 4A

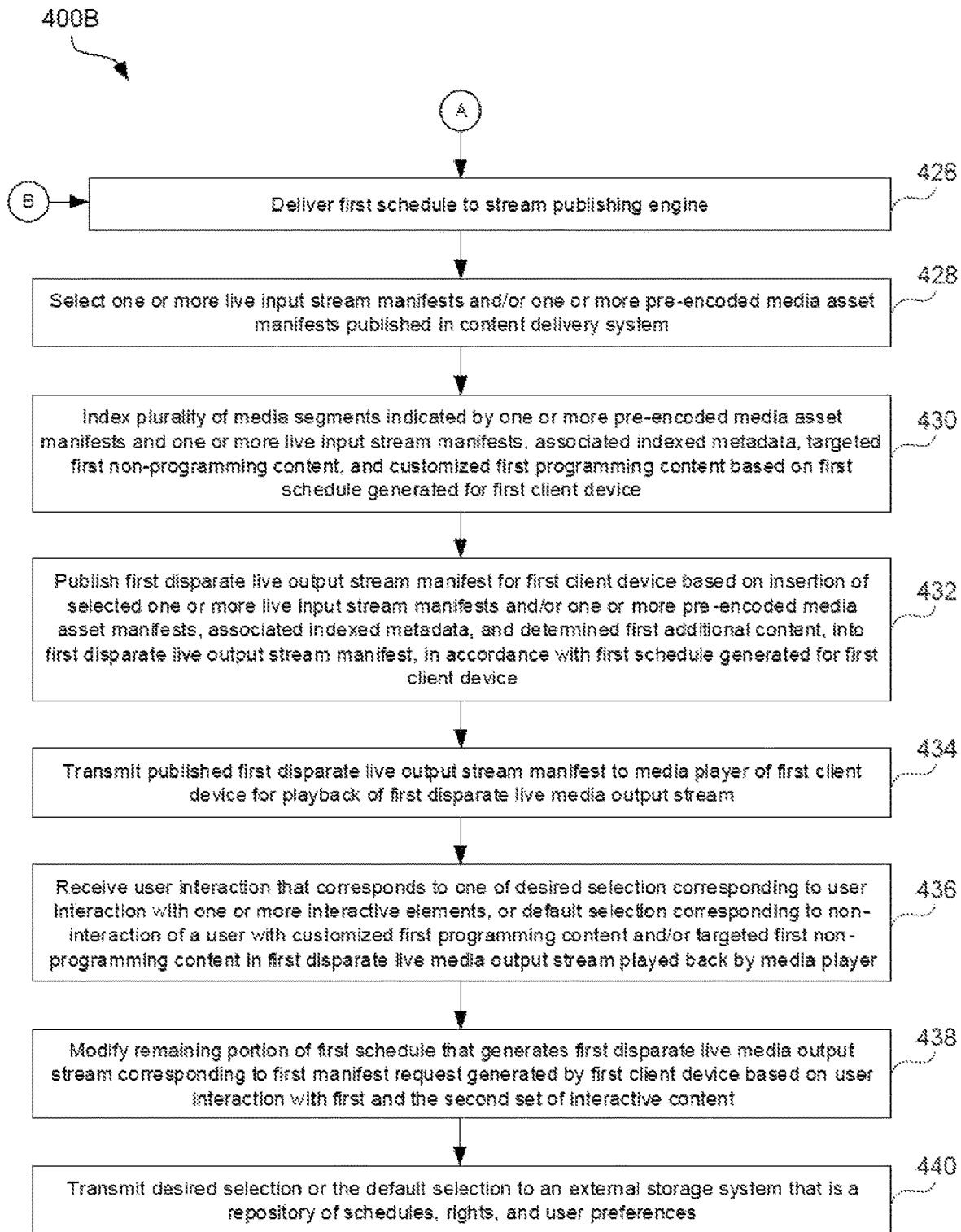


FIG. 4B

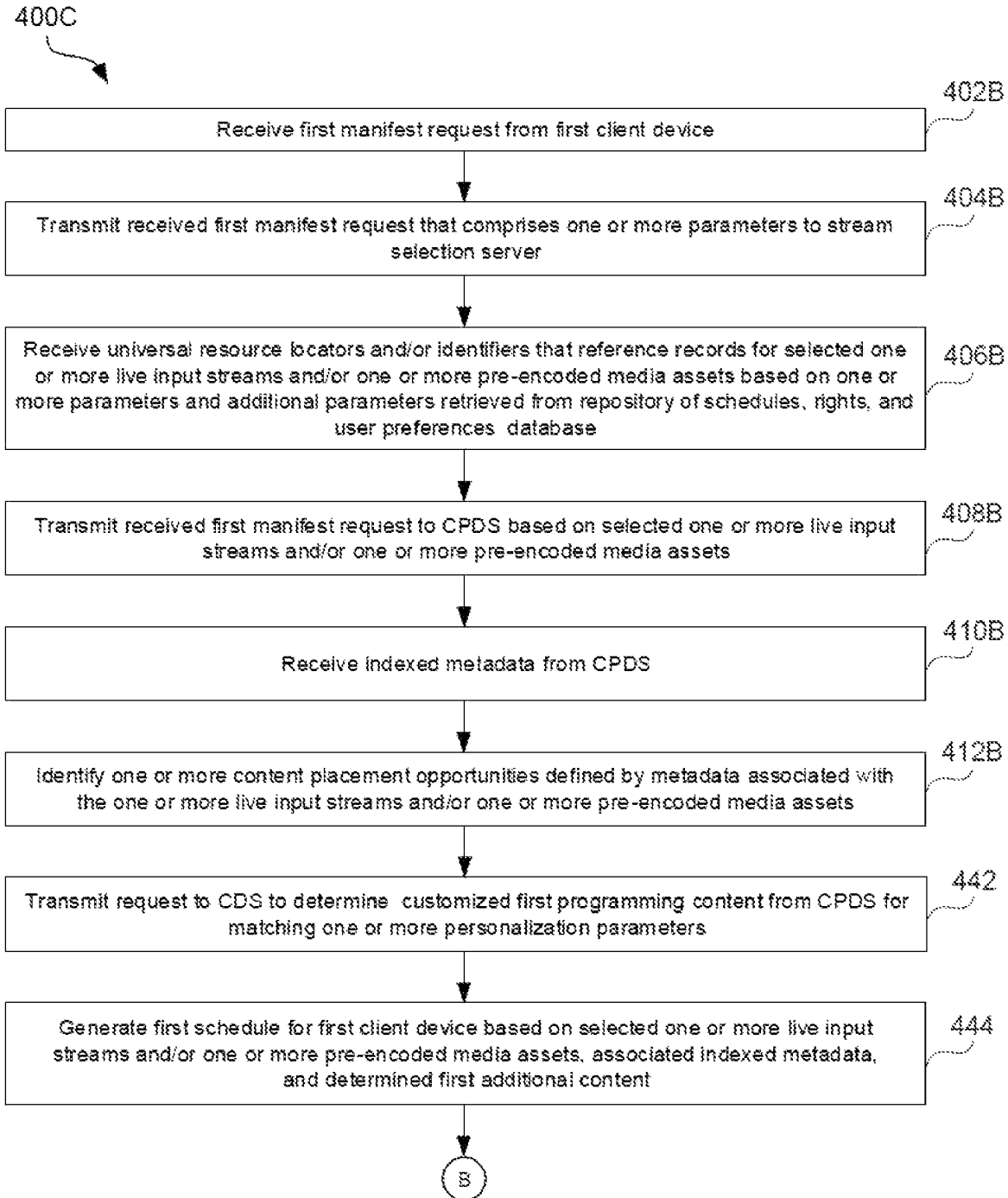


FIG. 4C

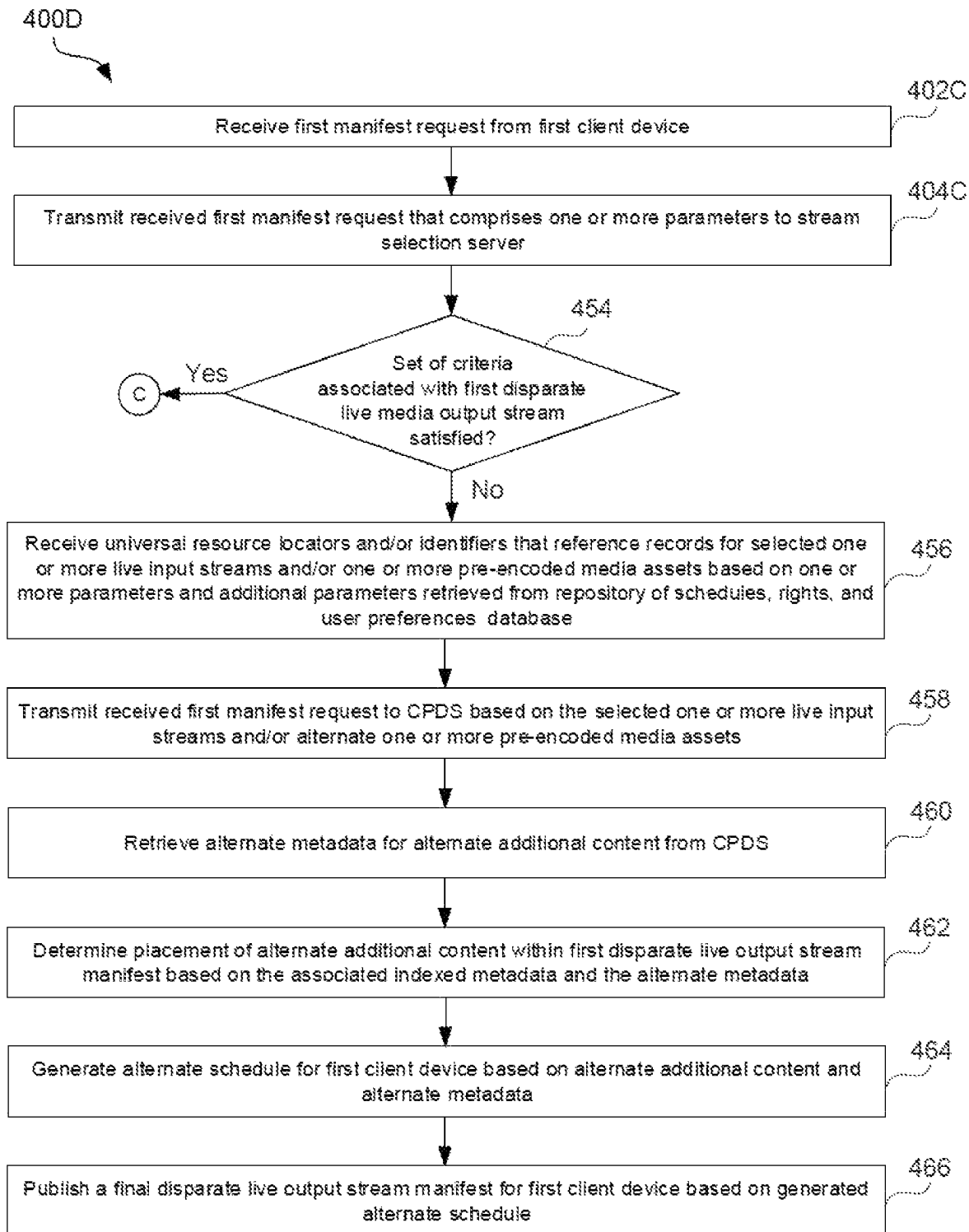


FIG. 4D

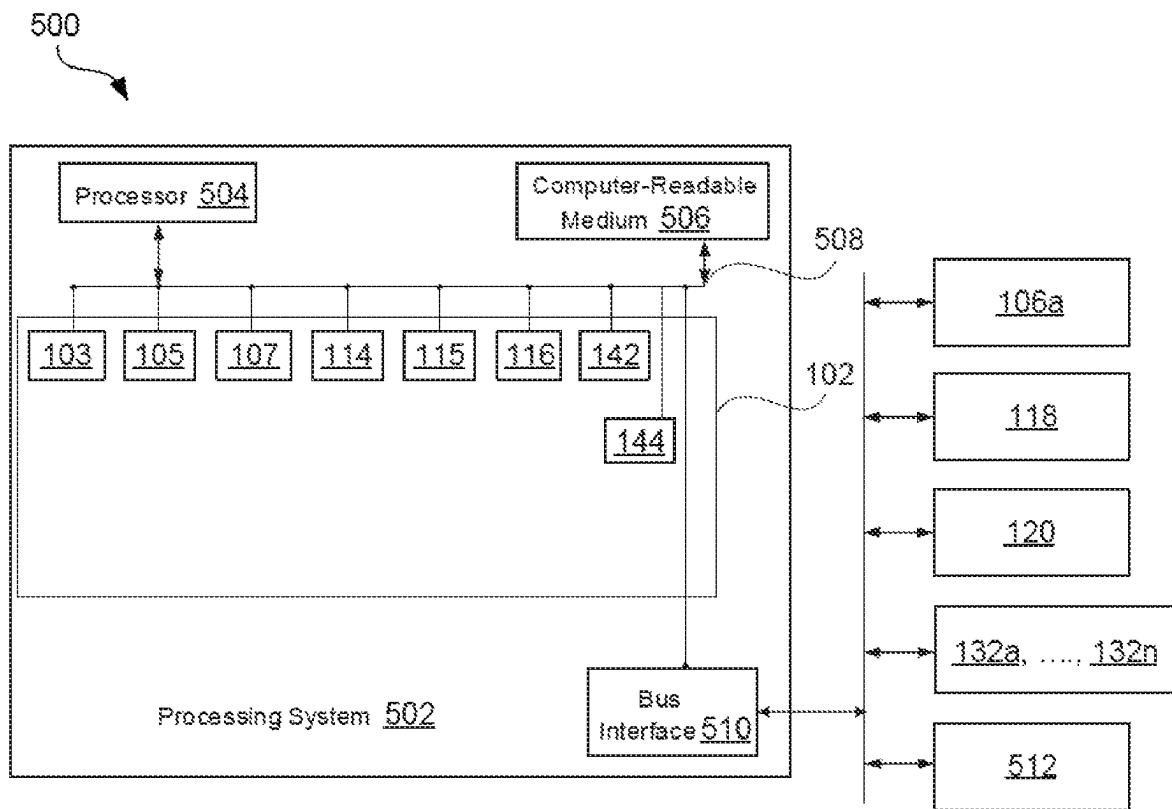


FIG. 5

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**PUBLISHING A DISPARATE PER-CLIENT
LIVE MEDIA OUTPUT STREAM BASED ON
DYNAMIC INSERTION OF TARGETED
NON-PROGRAMMING CONTENT AND
CUSTOMIZED PROGRAMMING CONTENT**

**CROSS-REFERENCE TO RELATED
APPLICATIONS/INCORPORATION BY
REFERENCE**

This Patent Application is a Continuation of U.S. patent application Ser. No. 17/094,319, filed on Nov. 10, 2020, which is further a Continuation of U.S. Pat. No. 10,965,967, issued on Mar. 30, 2021, which is a Continuation-in-part of U.S. Pat. No. 11,134,309, issued on Sep. 28, 2021, and claims priority to, and the benefit from U.S. Provisional Application Ser. No. 62/699,131, filed on Jul. 17, 2018.

This application also makes reference to:
U.S. Pat. No. 11,051,061, issued on Jun. 29, 2021;
U.S. Pat. No. 11,051,074, issued on Jun. 29, 2021;
U.S. Pat. No. 10,992,973, issued on Apr. 27, 2021;
U.S. Pat. No. 10,924,804, issued on Feb. 16, 2021; and
U.S. Pat. No. 10,856,016, issued on Dec. 01, 2020.

Each of the above referenced patent applications is hereby incorporated herein by reference in its entirety.

FIELD OF TECHNOLOGY

Certain embodiments of the disclosure relate to a television content packaging and distribution system. More specifically, certain embodiments of the disclosure relate to a method and system for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content.

BACKGROUND

Recent advancements in the field of television content packaging and distribution systems have led to a massive development of numerous technologies and broadcasting platforms that are revolutionizing the way consumer devices access and playout media content. Usually, broadcasting platforms refer to the types of networks that are used to deliver the media content to the consumers. Currently, the broadcasting platforms, such as analog terrestrial broadcast, digital terrestrial broadcast, direct-to-home satellite broadcast, cable, Internet Protocol (IP), and over-the-top television (OTT), compete and strive to increase their appeal by gaining and retaining the audience viewing the media content.

Modern streaming protocols, such as HTTP Live Streaming (HLS) and Dynamic Adaptive Streaming over HTTP (DASH), are implemented to support streaming of various live content services, such as DIRECTV NOWSM, SLING TVSM and PLAYSTAYIONTM VUE, to consumer devices. Due to dissemination of such modern streaming protocols in the television, radio, and broadcasting sector, it is evident that the success of broadcasting will be dependent on the ability of the network provider to gain access to the content that consumers demand, and to differentiate their offering from that of incumbent broadcasters or find breakthrough modes of media content delivery.

Existing systems for Server-Side Ad Insertion (SSAI) support live streaming and make decisions to insert non-programming content in near real-time. However, for On-Demand streaming, such decisions to insert non-programming content are made upfront. Accordingly, amount of control

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the user can excerpt to influence what non-programming content is shown may get limited.

Further limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of such systems with some aspects of the present disclosure as set forth in the remainder of the present application with reference to the drawings.

BRIEF SUMMARY OF THE DISCLOSURE

Systems and/or methods are provided for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content, substantially as shown in and/or described in connection with at least one of the figures, as set forth more completely in the claims.

These and other advantages, aspects and novel features of the present disclosure, as well as details of an illustrated embodiment thereof, will be more fully understood from the following description and drawings.

**BRIEF DESCRIPTION OF SEVERAL VIEWS OF
THE DRAWINGS**

FIG. 1A is a block diagram that illustrates an exemplary system for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content, in accordance with an exemplary embodiment of the disclosure.

FIG. 1B is a block diagram that illustrates an exemplary dynamic ad/content insertion system (DACIS) for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content, in accordance with an exemplary embodiment of the disclosure.

FIG. 2 illustrates segmentation of live input streams and pre-encoded media assets for a first programming schedule for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content by the DACIS of FIG. 1B, in accordance with an exemplary embodiment of the disclosure.

FIG. 3 illustrates an exemplary scenario associated with publishing a disparate live media output stream by the DACIS of FIG. 1B, in accordance with an exemplary embodiment of the disclosure.

FIGS. 4A to 4D collectively depict a flowchart illustrating exemplary operations for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content by the DACIS of FIG. 1B, in accordance with an exemplary embodiment of the disclosure.

FIG. 5 is a conceptual diagram illustrating an example of a hardware implementation for the DACIS employing a processing system for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content, in accordance with an exemplary embodiment of the disclosure.

**DETAILED DESCRIPTION OF THE
DISCLOSURE**

Certain embodiments of the disclosure may be found in a method and system for publishing a disparate per-client live media output stream based on dynamic insertion of targeted

non-programming content and customized programming content. Various embodiments of the disclosure provide a method and system that not only provide live channel offerings in cost-effective manner but also provide enhanced, intelligent, and personalized viewer experience to increase their appeal by retaining the audience viewing the media content.

Modern streaming protocols, such as HLS and DASH, break media content into numerous small media content segments, typically less than 10 seconds in length. Further, the modern streaming protocols implement manifests that instruct a media player on what media content segment to retrieve and play next. The manifest may enlist the media segments that make up the full length of the media asset. The manifest may include information, based on which the media player at a consumer device may be able to determine the media segments. The manifest and/or media content segment may also include and/or specify additional information to facilitate a media player to transition smoothly between media content streams from different sources. The manifest may be used for creating a playlist of multiple media content files, or for interrupting media content with advertising and then resuming the media content.

Such modern streaming protocols support video-on-demand (VOD) assets and live content as well. The VOD assets prepared for distribution, for example Internet distribution, may have a sequence of short duration segments added to a manifest. The short duration segments may be separate physical files or pointers (real or to be calculated) to the short media content segments inside a larger file. On the other hand, in case of live content, new short content media segments are made available as soon as they are created. In some protocols, each new segment is added to a manifest while in others the media player is provided with information that may be utilized to calculate what the next live segment will be. In the latter case, a signal in the media content itself may be utilized to inform the player when to re-inspect or check the manifest for a change in media content. In live streaming, delivery of live content is supported by making available each new short media content segments as soon as such media content segments are generated. In some protocols, new media content segments may be added to the manifest, while in others, the media player calculates necessary information about the next live media content segments.

In accordance with various embodiments of the disclosure, a system is provided for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content. One or more processors in the system may be configured to receive a first manifest request from a first client device, wherein the first manifest request comprises one or more parameters. The one or more processors in the system may be further configured to determine a first additional content that comprises a customized first programming content and a targeted first non-programming content for the first client device based on the one or more parameters and the associated indexed metadata. The one or more processors in the system may be further configured to generate a first programming schedule for the first client device based on the selected one or more live input streams and/or the one or more pre-encoded media assets, the associated indexed metadata, and the determined first additional content. Accordingly, one or more live input stream manifests and/or one or more pre-encoded media asset manifests published in a content delivery network and associated indexed metadata may be selected based on the

one or more parameters. The one or more processors in the system may be further configured to publish a first disparate live media output stream manifest for the first client device based on insertion of the selected one or more live input stream manifests and/or the one or more pre-encoded media asset manifests, the associated indexed metadata, and the determined first additional content in accordance with the first programming schedule generated for the first client device.

FIG. 1A is a block diagram that illustrates an exemplary system for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content, in accordance with an exemplary embodiment of the disclosure. Referring to FIG. 1A, the system 100, comprises a media packaging and distribution system 102 that is communicatively coupled to published data sources 104a, . . . , 104n, Ad decisioning servers 106a, . . . , 106n, via a network 108 and/or other programmatic means. There are shown consumer devices 110a, . . . , 110n that are communicatively coupled to the network 108. The media packaging and distribution system 102 may comprise at least a DACIS 103, a Content Decisioning Service (CDS) 105, a Content Packaging and Distribution System (CPDS) 107, a stream selection service 142, and a repository of schedules, rights, and user preferences database 144. There are also shown source devices 118 communicatively coupled to the media packaging and distribution system 102 through the network 108. An external data source 120 is also provided, which is communicatively coupled to the media packaging and distribution system 102 through the network 108.

The media packaging and distribution system 102 may comprise suitable logic, circuitry, and interfaces that may be configured to execute code that handles media content comprising audio, video, images, metadata, manifests, and/or other data (embedded and/or externally referenced). The media content may include a video, an audio, a combination of audio and video presentations, and/or embedded or externally referenced metadata, a combination of multiple-audio, multiple-video, and/or embedded or externally referenced metadata. Accordingly, the media packaging and distribution system 102 publishes a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content.

In this regard, the media packaging and distribution system 102 may provide video programming services to viewers, usually for a subscription fee (such as pay television). The media packaging and distribution system 102 also handles distribution, for example, multicasting, unicasting, broadcasting, streaming, for one or more channels to be viewed on one or more of the plurality of consumer devices 110a, . . . , 110n.

The media packaging and distribution system 102 may be operated by an entity related to handling or distribution of media content, for example, a broadcast provider or operator, or a network provider or network operator. The entity related to handling or distribution of media content may also be referred to as a content owner, a distributor, a syndicator, a re-distributor, a content aggregator, a search, discovery, or cataloging service provider, or any other entity actively or passively involved with the distribution, cataloging, or referencing of complete or partial presentations of media content. Throughout this document, the terms broadcast provider or broadcast operator, and network provider or network operator may be utilized to refer to the entity related to handling or distribution of media content, interchange-

ably. The broadcast provider may handle a single channel or a plurality of channels, or one or more networks. The broadcast provider may be configured to distribute content via one or more platforms, for example, traditional over-the-air broadcast channels, radio, cable television networks, satellite communication networks, the Internet, and/or other content delivery networks (CDNs). In this regard, the broadcast provider may be configured to execute code that communicates linear video feeds (also referred to as a network television feed or broadcast feed) to the media packaging and distribution system **102**. In a broadcast chain, the broadcast provider may receive actual content, for example, from a production studio, in one or more source formats. Examples of the one or more source formats may include, but are not limited to a tape file, or a live feed that may be further converted to a serial digital interface (SDI) video interface and/or on a high-definition SDI (HD-SDI) video interface for processing and playout. The broadcast provider may further process the content, such as insertion of graphics, closed captions, preparation of programming schedule, insertion of triggers, and the like, and provide final delivery by a broadcasting apparatus. The communicated linear video feed and the playout schedule may correspond to a channel, such as CNN channel that is broadcast to the media packaging and distribution system **102**, via a communication network. The linear video feed may be broadcasted as a multi-program transport stream (MPTS) (also referred to as a live video feed) to the media packaging and distribution system **102**, via the network **108**. The broadcast provider may be owned by (or associated to) a broadcast provider or operator, a network provider or operator, or a content provider or operator.

The media packaging and distribution system **102** may receive the MPTS, which includes the signaling content and metadata, from the broadcast provider based on, for example, current society of cable telecommunication engineers (SCTE) standards (SCTE-35 and SCTE-224) to control web and regional blackouts, network end of day switching, and advertisement insertion. For example, the media packaging and distribution system **102** may be signaled for various blackout types with in-band SCTE-35 message. Further, the media packaging and distribution system **102** may receive program metadata that specifies certain events or operations, such as, for example, when to blackout shows. Examples of legacy distribution system that may be benefitted from the media packaging and distribution system **102** may include direct-broadcast satellite (DBS) providers, cable television (CATV) systems, and other wireline video providers and competitive local exchange carriers (CLECs) using, for example, IPTV.

The DACIS **103** may comprise suitable logic, circuitry, and interfaces that may be configured to leverage one-to-one scale of traditional SSAI systems to support custom programming content choices and not just targeted non-programming content. For example when the user selects to join a live stream, a content decisioning system, such as the CDS **105** could determine that, instead of joining the live stream for the last few minutes of a program prior to the start of the intended program, the user should instead be shown content more relevant to the user so as to prevent them from tuning away, e.g. show a personalized set of basketball highlights and ads to a user who likely joined the stream to watch a basketball game that is coming on next. The DACIS **103** replaces original content in the live stream prior to game start with user relevant content, such that upon its conclusion, the user is seamlessly presented the game.

The DACIS **103** may be further configured to, via a programming schedule or tag indicator in a disparate live media output stream manifest, be notified of one or more content graphical treatment opportunities within the media content. The DACIS **103** may be further configured to make required non-programming content calls on behalf of the plurality of consumer devices **110a**, . . . , **110n**. Accordingly, the DACIS **103** may provide the plurality of consumer devices **110a**, . . . , **110n** with information needed to execute the graphical treatment graphical content via a secure out-of-band channel between the DACIS **103** and the plurality of consumer devices **110a**, . . . , **110n**. In accordance with an embodiment, the DACIS **103** may be configured to include not showing non-programming content that a user of a consumer device may elected to skip or rated poorly in an earlier non-programming content break. Further, the DACIS **103** may enable the user to skip non-programming content as the user interacted with a previous non-programming content or made a purchase and the advertiser elected to sponsor the remainder of the programming content.

In accordance with an embodiment, the DACIS **103** may be configured to provide seamless failover between redundant disparate live media output streams for large events, thus improving reliability. In certain instances, some of the plurality of consumer devices **110a**, . . . , **110n** may support a primary and backup disparate live media output streams and are able to fail between them. In other instances, others of the plurality of consumer devices **110a**, . . . , **110n** may not support the primary and backup disparate live media output streams. In such instances, the consumer devices may attempt to join an alternative disparate live media output stream after occurrence of an event such as a device failure or crash. For such consumer devices, the DACIS **103** may be configured to monitor both the primary and backup disparate live media output streams, and if there is a failure, write the alternative disparate live media output stream into the disparate live media output stream manifest.

Each of the plurality of published data sources **104a**, . . . , **104n** may be coupled to one or more television networks and may comprise suitable logic, circuitry, and interfaces that may be configured to execute code that provides actual audiences for programs that were distributed. As illustrated in FIG. 1A, the plurality of published data sources **104a**, . . . , **104n** are coupled to the media packaging and distribution system **102** via the network **108** and configured to monitor audience drift to or away from a tuned channel airing a live media output stream. The plurality of published data sources **104a**, . . . , **104n** may provide actual audiences for programs to the indexing and storage system **116**. An exemplary published data source may be Nielsen. Nielsen has the capability to determine when a viewer watches the same set of media items, for example, advertisements and/or promotional content, in programming data, such as an episode, in a live video feed within 3 days of original airing, and provide Nielsen "C3" credit. Another exemplary published data source may be a published database that provides ratings for a media item, such as gross rating point (GRP). The GRP is advertising or promotion impact measure for each advertising and/or promotional campaigns, known in the art. Other exemplary published data sources may also include direct/indirect access to other public or private data sources (persistent or ephemeral) through programmatic means.

The CDS **105** may comprise suitable logic, circuitry, and interfaces that may be configured to determine, upon request, which programming content (or partial programming content) to deliver back to the plurality of consumer devices **110a**, . . . , **110n**. Thus, the CDS **105** may be

configured to insert additional content (including non-programming content) or replace existing content, according to one or more parameters provided by the plurality of consumer devices **110a**, . . . , **110n** and data stored in various external systems and/or databases.

The Ad decisioning servers **106a**, . . . , **106n** may comprise suitable logic, circuitry, and interfaces that may be configured to implement at least an advertisement decisioning component that may be used during a real-time content or advertisement placement activity, for example during dynamic ad insertion. The Ad decisioning servers **106a**, . . . , **106n** may further determine ad-load opportunity, based on targeting data from schedules, rights, and user preferences database.

For example, commercial or non-commercial advertisements may be dynamically inserted within program segments of the live input streams based on the detected upcoming indicator, such as an inbound trigger, a signaling point, and/or a signal in a pre-encoded media asset and/or a live input stream by the Ad decisioning servers **106a**, . . . , **106n**.

In an embodiment, the DACIS **103** may be generalized as a proxy between the plurality of consumer devices **110a**, . . . , **110n** and the Ad decisioning servers **106a**, . . . , **106n**. In one implementation, a request is sent from a consumer device to the DACIS **103**. The DACIS **103** may call on of the Ad decisioning servers **106a**, . . . , **106n** to determine a set of non-programming content that may be inserted into the disparate live media output stream.

The CPDS **107** may comprise suitable logic, circuitry, and interfaces that may be configured to index programming content, which is prepared for usage by a system, such as the stream publishing engine **114**. The CPDS **107** may further define metadata detailing various facets of the programming and/or non-programming content including duration, known locations and opportunities for programming and/or non-programming content insertion/replacement.

The network **108** may be any kind of network, or a combination of various networks, and it is shown illustrating exemplary communication that may occur between the Ad decisioning servers **106a**, . . . , **106n** and the media packaging and distribution system **102**. For example, the network **108** may comprise one or more of a cable television network, the Internet, a satellite communication network, a wide area network (WAN), a medium area network (MAN), and a local area network (LAN). Although a network **108** is shown, the disclosure is not limited in this regard; accordingly, other exemplary modes may comprise uni-directional or bi-directional distribution, such as packet-radio, satellite. Furthermore, the network **108** is an exemplary embodiment of a distribution system.

The consumer devices **110a**, . . . , **110n** may refer to end-user devices or consumption devices where the content is played to be consumed by a user. The number of impressions of a media item, such as an advertisement and/or promotional media, on such plurality of consumer devices **110a**, . . . , **110n** determines the advertising impact or promotion impact and number of actual audiences achieved during campaigns. Examples of the consumer devices **110a**, . . . , **110n** may include, but are not limited to, connected TVs, connected TV with paired and/or connected devices (e.g., HDMI sticks, tablets), personal computer, smartphone, tablet, OTT set-top, or hybrid set-top, and second screen devices such as smartphones, tablets, game consoles, personal computers, set-top boxes, and embedded devices. The consumer devices **110a**, . . . , **110n** may further include process/system that may process the output for any

means, regardless of the capability or intent to decode for media presentation, and on which the consumer may launch a web page, a web application, or a web service to view media content.

The source devices **118** may comprise suitable logic, circuitry, and interfaces that may be configured to communicate a live media feed or live input streams of a channel, such as an existing channel, to the media packaging and distribution system **102**. In accordance with an embodiment, the live input streams of the channel may correspond to a broadcast feed. The source device **118** may be communicatively coupled to the network **108**.

The external data source **120** may comprise suitable logic, circuitry, and interfaces that may be configured to execute code that handles retrieval and storage of audience data that corresponds to subscribers of the plurality of consumer devices **110a**, . . . , **110n**. The audience data may include demographics data, audience targeting data, trending data, device type data, device platform data, and content recognition-based data, such as automatic content recognition (ACR)-based data. The trending data may comprise information on what's trending in the social networks (or platforms), such as Twitter®, Facebook®, and the like. The trending data also comprises information on what's trending based on social engagement, such as number of likes or votes to a particular media item, or number of people watching a particular media item. The trending data may indicate an aggregate interest level of a number of users in the social networks for a particular media item. For example, a thousand or more shares, or likes by a number of users may indicate a highly popular media item.

The stream selection service **142** may comprise suitable logic, circuitry, and interfaces that may be configured to execute code that may be configured to provide a consumer device, for example, the consumer device **110a**, requesting to view the disparate live media output stream with a correct variant of disparate live media output stream, based on the geolocation and identification of the consumer device **110a**, along with data retrieved from the repository of schedules, rights, and user preferences database **144**.

The repository of schedules, rights, and user preferences database **144** may comprise suitable logic, circuitry, and interfaces that may be configured to store the schedules for all source feeds, availability rights for all the content in the schedules, regional blackout zones for the various sports leagues, predefined location-based viewing preferences, individual client viewing preferences, and any viewing or transition rules provided by the stream owner operator.

In operation, in accordance with an aspect of the disclosure, upon receiving a first manifest request from a consumer device, the DACIS **103** may be configured to publish a manifest of a first disparate live media output stream for the consumer device. The first disparate live media output stream may be chosen by the DACIS **103**, at the time of the first manifest request, to best target a consumer device, according to a plurality of client-specific parameters. The plurality of client-specific parameters may comprise user preferences and identifiers, consumer device preferences and identifiers, and one or more rules/rights governed by geolocation data and current position of playback of a first disparate live media output stream at the consumer device. The first disparate live media output stream manifest may be generated, at the time of the first manifest request, to include additional content, such as the targeted non-programming content and the customized programming content. Examples of the targeted non-programming content may include, but are not limited to, advertisements, personalized (per-user)

advertisements, video advertisements, and graphical treatment (such as overlays). The targeted non-programming content may further include first interactive elements, such as an option to purchase goods or content, and second interactive elements. The second interactive elements may further affect the generation of the first disparate live media output stream manifest by providing various options. For example, a first option may be to skip a given ad, a second option may be to skip future ads for remainder of the playback of the first disparate live media output stream, a third option may be to choose one or more ads of a given category, or a fourth option may be to watch all ads immediately to avoid ads for the remainder of the playback. Examples of the customized programming content may include, but are not limited to, promotional content, short-form content, or alternate content (content replacing that of the requested first disparate live media output stream).

The DACIS 103 may further receive metadata associated with the content and additional content for the first disparate live media output stream from the CPDS 107. The metadata may include, for example, ad break locations, graphical treatment marker/triggers, SCTE35 markers, content duration, personalized content opportunities, and one or more decision point locations.

In accordance with an embodiment, the DACIS 103 may be configured to convert a pre-encoded media asset to the first disparate live media output stream to facilitate one or more subsequent modifications on the first disparate live media output stream. The one or more subsequent modifications may correspond to a user selection, a user preference, a change in a first programming schedule, or a time or geolocation-based rule. In accordance with an embodiment, the DACIS 103 may be configured to perform another conversion of remainder of the first disparate live media output stream into the one or more pre-encoded media assets to facilitate download of the one or more pre-encoded media assets at the consumer device and mitigate dependency on the DACIS 103 for playback of remaining portion.

In accordance with another aspect of the disclosure, upon receiving the first manifest request, a manifest of the first disparate live media output stream may be altered to include additional or alternate content, apart from the content originally scheduled for playback. The DACIS 103 may be configured to determine which content to include/replace by utilizing one or more parameters. The one or more parameters may include, but not limited to, URL requested for playback, and an identifier referencing a record for an existing first disparate live media output stream or an external connected system, such as the CPDS 107. The one or more parameters may include a plurality of client-specific parameters that may comprise user preferences and identifiers, consumer device preferences and identifiers, and one or more rules/rights governed by geolocation data and current position of playback of a first disparate live media output stream at the consumer device. The one or more parameters may include a plurality of client-specified attributes derived from user interaction with the consumer device. The plurality of client-specified attributes may include, for example, a preference for a given type and/or category (or categories) of programming content, a possible time constraint or duration to fill with content. The time constraint or duration to fill with content may be determined by user preferences from an external system, such as the repository of schedules, rights, and user preferences database 144, a range defined in the first client manifest request, schedule tolerances, as defined in the repository of schedules, rights, and user preferences database 144, device

preferences/identifiers declared in the first manifest request, and a prior consumer request, or found in the repository of schedules, rights, and user preferences database 144. The time constraint or duration to fill with content may be further determined by geolocation information declared in the first manifest request or found in the first manifest request, or external database, such as the repository of schedules, rights, and user preferences database 144, or external systems, such as a content recommendation engine. The one or more parameters may further include rules, rights, and schedule data stored in the DACIS 103 or an adjacent, external system—such as the repository of schedules, rights, and user preferences database 144.

In accordance with an embodiment, the DACIS 103 may determine where to place new or alternate content within the requested disparate live media output stream. The DACIS 103 may utilize metadata, such as ad break locations, graphical treatment markers/triggers, SCTE35 markers, content duration, and one or more decision point locations. The metadata may be sourced from a system that has indexed metadata for the asset, such as the CPDS 107. The DACIS 103 may generate a new schedule from the determined alterations in the first disparate live media output stream. The DACIS 103 may transmit resulting schedule to the stream publishing engine 114 to generate a final disparate live media output stream. The final disparate live media output stream may be delivered back to originating client. The stream publishing engine 114 may generate the first disparate live media output stream manifest based on one of a pre-defined conversion modes. The pre-defined conversion modes may correspond to pre-encoded media assets to live stream mode, pre-encoded media assets to live stream mode with scalable architecture, a live stream to live stream mode, and a mixed mode corresponding to switching between pre-encoded media assets and live streams.

The final disparate live media output stream may contain one or more decision points, which, if configured, may be presented by the consumer device to the user. In this regard, the final disparate live media output stream may be delivered as a live stream between the decision points whether or not the programming content selected by a user is pre-encoded media asset or a live input stream. At the one or more decision points, the user interaction may correspond to one of a desired selection corresponding to the user interaction with one or more interactive elements, or a default selection corresponding to non-interaction of a user with the customized first programming content and/or the targeted first non-programming content in the first disparate live media output stream played back by the media player. The desired selection or the default selection may be transmitted to an external storage system that is the repository of schedules, rights, and user preferences database 144.

In accordance with an embodiment, at the one or more decision points, the one or more interactive elements may facilitate exclusion of the targeted first non-programming content or the customized first programming content and replacement by default content. The one or more interactive elements may further facilitate selection of alternate customized first programming content, selection of a subsequent second programming content, approval or disapproval of the selected customized first programming content, and/or exclusion of subsequent second non-programming content for a remaining portion of the first disparate live media output stream played back by the media player. The one or more interactive elements may further facilitate selection of one or more targeted first non-programming content of a specific category, or viewing of some or all of the targeted

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first non-programming content immediately to avoid some or all of the targeted first non-programming content for the remaining portion of the first disparate live media output stream played back by the media player.

In accordance with another aspect of the disclosure, the DACIS 103, upon the first manifest request, may seamlessly transition the first additional content to an alternate additional content according to determination made by the stream selection service 142, based on one or more transition parameters. The one or more transition parameters may comprise one or more parameters from the first manifest request, current state of the first disparate live media output stream manifest determined based on accessibility, regular update, and suitable encoding, digital rights management, and compatibility with the first client device, rules provided by a stream owner operator, and user preferences defined in the repository of schedules, rights, and user preferences database 144.

The DACIS 103 may retrieve alternate metadata for the alternate additional content from the CPDS 107. The alternate metadata may indicate a location to seamlessly transition from the first additional content to the alternate additional content, and may include ad break locations, graphical treatment markers/triggers, SCTE35 markers/triggers, content duration, and/or one or more decision points. The DACIS 103 may be configured to determine placement of the alternate additional content within the first disparate live media output stream manifest based on the associated indexed metadata and the alternate metadata. The DACIS 103 may generate an alternate programming schedule for the first client device based on the alternate additional content and the alternate metadata. The DACIS 103 may transmit the generated alternate programming schedule to the stream publishing engine 114. The stream publishing engine 114 may generate a final disparate live output stream manifest for the consumer device.

In accordance with an embodiment, the DACIS 103 may revoke the first disparate live media output stream manifest published for the first consumer device based on an identifier primitive associated with the first disparate live media output stream of the first consumer device in an instance in which a media player of the first consumer device is determined to be a plagiarized media player with unauthorized access to content. In such embodiment, the first disparate live media output stream may include at least one unique identifier inserted by the DACIS 103.

FIG. 1B is a block diagram that illustrates an exemplary DACIS for publishing an updated disparate live media output stream in mixed mode based on user selection, in accordance with an exemplary embodiment of the disclosure. Referring to FIG. 1B, the DACIS 103 comprises a first programming schedule 111, an alternate programming schedule 113, a stream publishing engine 114, a personalized experience manager (PEM) 115, and an indexing and storage system 116. FIG. 1B further illustrates a content delivery system 130, which is an example of the network 108, client devices 132a, . . . , 132n, which correspond to the plurality of consumer devices 110a, . . . , 110n. There is further shown a per-client disparate live media output stream 117a, . . . , 117n for corresponding client device from the client devices 132a, . . . , 132n.

In some embodiments of the disclosure, the stream publishing engine 114, the PEM 115, and the indexing and storage system 116 may be integrated to form an integrated system, as illustrated in FIG. 1B. In some embodiments of the disclosure, as shown, the stream publishing engine 114, the PEM 115, and the indexing and storage system 116 may

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be distinct. In this regard, the PEM 115 may be implemented external to the DACIS 103 without loss of generality. Other separation and/or combination of the various entities of the exemplary media packaging and distribution system 102 illustrated in FIG. 1B may be done without departing from the spirit and scope of the various embodiments of the disclosure. In an embodiment, the implementation of the DACIS 103 may be on the server-side. In another embodiment, the implementation of the DACIS 103 may be on the client-side.

The first programming schedule 111 may correspond to an instruction set for a disparate live media output stream for a corresponding client device. The first programming schedule 111 may inform the stream publishing engine 114 about pre-encoded media assets and live input streams as well as when and how to switch between the various pre-encoded media assets and live input streams. The first programming schedule 111 may also support defining break durations for mid roll ads, break locations, and durations in the pre-encoded media asset and live input stream switches.

The alternate programming schedule 113 may correspond to an instruction set for an updated or alternate disparate live media output stream. The alternate programming schedule 113 may inform the stream publishing engine 114 about the alternate pre-encoded media asset and/or the alternate live input stream, and alternate additional content. Specifically, the alternate programming schedule 113 may indicate that when and how to switch between the various disparate live media output streams.

The stream publishing engine 114 in the DACIS 103 may be configured to generate disparate live media output stream manifests and variants of disparate live media output stream manifests. The stream publishing engine 114 may be configured to publish unique-to-client streaming manifests leveraging different indexes created by the indexing and storage system 116 from the various live input streams, pre-encoded media assets, targeted non-programming content and customized programming content based on a defined per-client schedule.

The PEM 115 in the DACIS 103 may be configured to personalize viewer experience of users by communicating with existing content decisioning systems, such as CDS 105, and executing/converting schedules provided, for example from the first programming schedule 111 to the alternate programming schedule 113. The PEM 115 in the DACIS 103 may be further configured to communicate with an ad server, such as the Ad decisioning server 106a, and stitching in targeted first non-programming content and customized first programming content, as required by the first programming schedule 111 and user preferences. The PEM 115 in the DACIS 103 may be further configured to generate a schedule for a disparate live media output stream of personalized content (i.e., clips or movies), communicating with the stream selection service 142 to switch or failover to different disparate live media input streams, and control overlays and other events triggered via the disparate live media output stream. In accordance with an embodiment, the stream selection service 142 may be used to switch between different output streams when a client device is requesting directly from the stream selection service 142. In accordance with other embodiment, the stream selection service 142 may be used to switch between different streams that may act as inputs for the PEM 115 and/or stream publishing engine 114 to generate a new output stream.

The indexing and storage system 116 in the DACIS 103 may be configured to ingest pre-encoded media assets, advertisement, and (continuously) live stream source mani-

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fest, indexes the media content segments, indexes one or more program indicators (such as program boundaries), non-programming indicators (such as ad break locations, overlay opportunities credits, and DRM systems supported, in the repository of schedules, rights, and user preferences database 144.

The per-client disparate live media output streams 117a, . . . , 117n may correspond to disparate live media output stream comprising live input streams and/or pre-encoded media assets to be sent back to a client device, upon request, that has been generated from a per-client schedule, such as the first programming schedule 111. Various media container formats of the live input streams and/or pre-encoded media assets may include, but are not limited to, transport stream (TS), fragmented MP4 (fMP4), Common Media Application Format (CMAF) and the like.

The content delivery system 130 may correspond to the network 108, described in FIG. 1. The content delivery system 130 may comprise networks configured for distributing media content to the plurality of client devices 132a, . . . , 132n. Generally, the term “content,” “metadata,” “media,” and similar words are used interchangeably to refer to any type of media—audio, videos, datacasts, music, text, images, graphics, articles, still photos, animated photos, moving photos, photo galleries, video galleries, infographics, maps, polls, guest biographies, tweets or other social media, blog posts, and/or the like. The content delivery system 130 may be configured to provide a plurality of disparate live media output streams to the plurality of client devices 132a, . . . , 132n (in case packaged content is available on the network 108) or from the content packaging and distribution system 107 (in case packaged content is not available on the network 108). The plurality of disparate live media output streams may be provided to the plurality of client devices 132a, . . . , 132n via, for example, a transport stream, a segmented streaming, a progressive download, or any other modes of distributing a multimedia presentation, such as via an over-the-air content delivery/distribution network, a linear content delivery/distribution network, a cable content delivery/distribution network, a satellite content delivery/distribution network, an Internet Protocol (IP) based content delivery/distribution network, and/or the like.

The client devices 132a, . . . , 132n may correspond to consumer devices 110a, . . . , 110n. In accordance with an embodiment, the client devices 132a, . . . , 132n may be content recognition (CR)-enabled devices, such as automatic content recognition enabled devices. The client devices 132a, . . . , 132n may be configured to communicate with the Ad decisioning server 106a, via the content delivery system 130, or a separate communication network.

The stream selection service 142 may comprise suitable logic, circuitry, and interfaces that may be configured to provide a client device, for example, the client device 132a, requesting to view a disparate live media output stream with a correct variant of disparate live media output stream, based on the geolocation and identification of the client device 132a, along with data retrieved from the repository of schedules, rights, and user preferences database 144. The stream selection service 142 may further receive user preferences of a user associated with the client device 132a, for example, to view the recommended/desired existing disparate live media output stream on the client device 132a. The stream selection service 142 may further store the received preferences in the repository of schedules, rights, and user preferences database 144, and also communicate with the PEM 115. The stream selection service 142 acts as an

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interface between the PEM 115 of the media packaging and distribution system 102 and the plurality of client devices 132a, . . . , 132n.

The repository of schedules, rights, and user preferences database 144 may comprise suitable logic, circuitry, and interfaces that may be configured to execute code to store content rights, user preferences, regional preferences, live schedules, and regional rights. For example, content rights may store availability and platform rights for live input streams in the first programming schedules, such as the first programming schedule 111, the user preferences may store preferences of individual user viewing preferences, the regional preferences may store regional viewing preferences, the live schedules may store the schedules for live input streams, and the regional rights may store regional blackout zones for the various sports leagues, for example. The repository of schedules, rights, and user preferences database 144 may further store data supplied from a stream owner/operator including requirements, preferences, such as pre-defined location-based viewing preferences, stream transition rules, and any required client data, such as service level and zip code.

In operation, in accordance with an aspect (regarding just-in-time insertion of non-programming content and/or programming content) of the disclosure, a media player in a consumer device, such as the consumer device 110a, may send a first manifest request to the DACIS 103 to receive a disparate live media output stream manifest to begin or continue playback. The first manifest request may comprise one or more parameters, such as universal resource locators and/or identifiers referencing records for one or more live input streams or one or more pre-encoded media assets in the CPDS 107, a plurality of client-specific parameters, and a plurality of client-specified attributes derived from a user interaction with the consumer device, such as the consumer device 110a. The plurality of client-specific parameters may comprise user preferences and identifiers, client device preferences and identifiers, and one or more rules governed by geolocation data and current position of playback of a first disparate live media output stream at the consumer device, such as the consumer device 110a. The plurality of client-specified attributes may comprise the user interaction with interactive content in a customized first programming content and a targeted first non-programming content, and a preference for a type and/or category of the targeted first non-programming content and/or the customized first programming content. The user interaction with the interactive content may comprises a selection to exclude the targeted first non-programming content and/or the customized first programming content, a selection to include a subset of the targeted first non-programming content and/or the customized first programming content, and a selection to include all of the targeted first non-programming content and/or the customized first programming content within one or more specified non-programming content locations.

In case the first manifest request is for a first disparate live media output stream, the PEM 115 transmits the one or more parameters in the first manifest request to the stream selection service 142. The stream selection service 142 may determine live input streams and/or pre-encoded media assets, based on the one or more parameters in the first manifest request and in repository of schedules, rights, and user preferences database 144. The stream selection service 142 may transmit the determined identifiers or URLs of the live input streams and/or pre-encoded media assets to the PEM 115. The PEM 115 may then transmit the first manifest request to the CPDS 107 to retrieve metadata for the

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determined live input streams and/or pre-encoded media assets. The metadata may include, for example ad break locations, overlay markers/triggers, SCTE35 markers, and content duration.

The PEM 115 of the DACIS 103 may further transmit the received first manifest request to the CPDS 107 based on the selected one or more live input streams and/or one or more pre-encoded media assets. The PEM 115 may receive the indexed metadata, for example non-programming indicators, such as ad break locations, graphical treatment indicators, such as overlay markers/triggers, programming indicators, such as SCTE35 markers and content duration, from the CPDS 107.

The PEM 115 may further identify the one or more content placement opportunities defined by the metadata associated with the one or more live input streams and/or one or more pre-encoded media assets. The PEM 115 may further receive additional content, such as targeted first non-programming content from the Ad decisioning server 106a, and customized first programming content to match one or more personalization parameters from the CDS 105.

The PEM 115 may be configured to generate the first programming schedule 111 for a first client device, such as the client device 132a, based on selected one or more live input streams and/or the one or more pre-encoded media assets, the associated indexed metadata, and the determined first additional content. The generated first programming schedule 111 may be delivered to the stream publishing engine 114. The stream publishing engine 114 may be configured to select the one or more live input stream manifests and/or the one or more pre-encoded media asset manifests published in the content delivery system 130. The indexing and storage system 116 may be configured to index the plurality of media segments indicated by the one or more pre-encoded media asset manifests and the one or more live input stream manifests, the associated indexed metadata, the targeted first non-programming content, and the customized first programming content may be indexed based on the first programming schedule 111 generated for the first client device, for example the client device 132a. The stream publishing engine 114 may be further configured to publish the first disparate live media output stream manifest for the first client device, for example the client device 132a. The DACIS 103 may transmit the published first disparate live media output stream manifest to the media player of the first client device for playback of the first disparate live media output stream, for example the disparate live media output stream 117a.

In accordance with an embodiment, the PEM 115 may receive a user interaction from the media player of the first client device, for example the client device 132a. The user interaction may correspond to one of a desired selection corresponding to the user interaction with one or more interactive elements, or a default selection corresponding to a non-interaction of the user with the customized first programming content and/or the targeted first non-programming content in the first disparate live media output stream 117a played back by the media player. Based on the user interaction with the interactive content, the PEM 115 of the DACIS 103 may be configured to modify the remaining portion of the first programming schedule 111 that generates the first disparate live media output stream 117a corresponding to the first manifest request generated by the first client device, such as the client device 132a.

In accordance with another aspect (regarding personalized insertion of playlist of non-programming content and/or programming content) of the disclosure, the DACIS 103

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may be configured to receive the first manifest request from the first client device, such as a client device 132a, to begin or continue playback of a first disparate live media output stream, for example the disparate live media output stream 117a. In such embodiment, in addition to the plurality of client-specific parameters described above, the plurality of client-specified attributes may comprise a preference for a given type and/or category (or categories) of targeted non-programming content or customized programming content, and a possible time constraint (duration) to fill with content. Thereafter, in similar manner as described above, the PEM 115 may transmit the one or more parameters in the first manifest request to the stream selection service 142. The stream selection service 142 may determine live input streams and/or pre-encoded media assets, based on the one or more parameters in the first manifest request and in repository of schedules, rights, and user preferences database 144. The stream selection service 142 may transmit the determined identifiers or URLs of the live input streams and/or pre-encoded media assets to the PEM 115. The PEM 115 may then transmit the first manifest request to the CPDS 107 to retrieve metadata for the determined live input streams and/or pre-encoded media assets. The metadata may include, for example, categories to which content is assigned to (for example, "basketball", "sports", "Knicks"), in addition to ad break locations, overlay markers/triggers, SCTE35 markers, and content duration.

The PEM 115 may identify one or more content placement opportunities defined by the metadata and may transmit a request to the CDS 105 to determine the customized first programming content from the CPDS 107 for matching one or more personalization parameters. Accordingly, the PEM 115 may generate the first programming schedule 111 defining locations and types of one or more decision points defined by the one or more content placement opportunities in the indexed metadata. The one or more decision points may define personalized content preferences, skipping of personalized content and moving to default content, and approval or disapproval of the personalized content. The remaining operations may be similar to the just-in-time insertion of non-programming content and/or programming content.

In accordance with another aspect (regarding stream failover) of the disclosure, the DACIS 103 may be configured to receive the first manifest request from the first client device, such as a client device 132a. The first manifest request may comprise one or more parameters, as described above. The PEM 115 may transmit the received first manifest request that comprises the one or more parameters to the stream selection service 142. The stream selection service 142 may determine if the set of criteria associated with the first disparate live media output stream is satisfied. The set of criteria may include accessibility of the first disparate live media output stream, update of the first disparate live media output stream and/or compatibility of the first disparate live media output stream having media and/or a manifest with the first manifest request. The stream selection service 142 may be further configured to determine the one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters in the first manifest request, and additional parameters retrieved from the repository of schedules, rights, and user preferences database 144. The stream selection service 142 may be further configured to determine the one or more live input streams and/or one or more pre-encoded media assets based on rules provided by a stream owner/operator (such as a regional blackout for the first client device leading to an alternative stream to

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watch) and user preferences (that exclude certain categories) defined in the repository of schedules, rights, and user preferences database **144**.

When the set of criteria associated with the first disparate live media output stream is not satisfied, the stream selection service **142** may select a pre-encoded asset indicated in the first manifest request to continue playback as the first disparate live media output stream. The PEM **115** may receive the universal resource locators and/or identifiers that reference records for the selected one or more live input streams or alternate one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**. The PEM **115** may further transmit the received first manifest request to the CPDS **107** based on the selected one or more live input streams or alternate one or more pre-encoded media assets. The PEM **115** may further receive the alternate metadata for the alternate additional content from the CPDS **107**. The metadata may include, for example, Ad break locations, overlay markers/triggers, SCTE35 markers/triggers, content duration, and one or more decision points. The alternate metadata may further include a location to transition from the first additional content to the alternate additional content. Thereafter, a placement of alternate additional content within the first disparate live media output stream manifest may be determined based on the associated indexed metadata and the alternate metadata. The PEM **115** may further generate the alternate programming schedule for the first client device, such as the client device **132a**, based on the alternate additional content and alternate metadata. Accordingly, the stream selection service **142** may publish the final disparate live output stream manifest for the first client device, such as the client device **132a**, based on the generated alternate programming schedule.

FIG. 2 illustrates segmentation of live input streams and pre-encoded media assets for the first programming schedule **111** or the alternate programming schedule **113** for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content by the DACIS **103** of FIG. 1B, in accordance with an exemplary embodiment of the disclosure. Referring to the exemplary arrangement of FIG. 2, there is shown a first live input stream **202**, a first pre-encoded media asset **204**, and a second live input stream **206**. There is also shown a targeted first non-programming content **208** placed after the first live input stream **202**, and a customized first programming content **210** placed after the first pre-encoded media asset **204**. The first live input stream **202** may be segmented into a first set of video segments **202a, 202b, 202c, . . . , 202n**. Similarly, the first pre-encoded media asset **204** and the second live input stream **206** may also be segmented into second set of video segments **204a, 204b, 204c, . . . , 204n**, and third set of video segments **206a, 206b, 206c, . . . , 206n** respectively. By way of example, the segmentation may be executed by a segmenting system (for example a live stream encoder/packager and/or a content encoder/packager (not shown)) during a preparation stage of the media assets. The encode stage may create various quality levels and the package stage segments the content into the short segments, and produces the correct format, such as TS, fMP4, or CMAF and encrypts the media content to prevent piracy. In accordance with an embodiment, the segments of the first set of video segments **202a, 202b, 202c, . . . , 202n**, the second set of video segments **204a, 204b, 204c, . . . , 204n**, and third set of video segments **206a, 206b, 206c, . . . , 206n**, may be segmented into consistent

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length, for example, 10 seconds segments. It may be advantageous to have a consistent and smaller file size of segments to be able to quickly push to the content delivery system **130**, and also for quick downloading by a media player at the end-user side, such as on the plurality of consumer devices **110a, . . . , 110n**.

It should be understood by those skilled in the art that various changes may be made and segments of different file sizes (or length) may be used without departure from the scope of the present disclosure. Further, other streaming protocols may require a different processing of media content. Thus, the scope of the disclosure should not be limited to the processing or preparation of media content to allow delivery using different delivery methods, streaming protocols, or distribution system, known in the art. Further, instead of the live input streams and pre-encoded media asset arranged, as shown, different arrangements per the first programming schedule **111** or the alternate programming schedule **113** may be possible with respect to interstitial content items, such as the targeted first non-programming content **208** and the customized first programming content **210**.

The insertion of the live input stream manifests, pre-encoded media asset manifests, the targeted first non-programming content **208** and the customized first programming content **210** may be done on-the-fly based on dynamic scheduling by the PEM **115** that generates the first programming schedule **111** or the alternate programming schedule **113**. The insertion may be driven by real time or near-real time content context analysis, user-selection on the consumer devices **110a, . . . , 110n**, or driven by external data received from the external data source **120**. The PEM **115** in association with the stream selection service **142** may be configured to insert live input streams, such as the first live input stream **202** and the second live input stream **206**, or pre-stored media assets, such as the first pre-encoded media asset **204**, the targeted first non-programming content **208** and the customized first programming content **210**, in an existing disparate live media output stream based on manipulation of a manifest the existing disparate live media output stream, such as an existing channel.

In accordance with an embodiment, each segment of the first set of video segments **202a, 202b, 202c, . . . , 202n**, the second set of video segments **204a, 204b, 204c, . . . , 204n**, and third set of video segments **206a, 206b, 206c, . . . , 206n**, may be further processed to be stored at various quality levels, and content encryption modes for the purposes of adaptive bitrate streaming and digital rights management, for example, the video segment **202a** may be stored in a plurality of quality levels, for example, high definition (HD), high dynamic range (HDR) video, or different quality levels in accordance with specified pixel resolutions, bitrates, frame rates, and/or sample frequencies. As each of the media content, such as **202** to **206**, are encoded, segmented, and stored with the plurality of quality levels in a media content master storage system. The media content may be re-used to create new channels, such as a new disparate live media output stream, without having to re-encode a selected live input stream or a pre-encoded media asset when a new disparate live media output stream is created using the live input streams or a pre-encoded media asset.

For the sake of brevity, and with reference to FIG. 2, there is shown an example of publishing the first or the updated (or alternate) disparate live media output streams based on dynamic insertion of targeted non-programming content and customized programming content by the DACIS **103** of FIG. 1B. It is to be understood that media packaging for different

delivery methods (such as analog terrestrial broadcast, digital terrestrial broadcast, direct-to-home satellite broadcast, cable, other Internet Protocol (IP)-based delivery methods, over-the-top television (OTT)), different streaming protocols, or distribution system, may be different. The media content may be prepared to be viewed one or more of the plurality of consumer devices **110a**, . . . , **110n**, based on at least the desired delivery method, delivery conditions, content protection requirements, to satisfy operational and technical requirements, as needed. The operational and technical requirements may include, but are not limited to, media encoding, media segmentation, programming schedule (or manifest) creation or manipulation requirements, desired media encryption, and/or metadata signaling requirements. For example, in certain scenarios and for certain media content delivery methods, network bandwidth, network conditions, or device-type where media content is to be consumed may not be variable or known in advance. In such a case, creating different quality levels for same media content may not be required. Further, based on different operational and technical requirements, publishing of disparate live media output stream may be different. The media content that is prepared and distributed may include both the programming content, such as long-form presentations, short-form presentations, news or sporting events; and non-programming content, such as paid advertisements, public service advertisements, or promotional material.

FIG. 3 illustrates an exemplary scenario associated with publishing a disparate live media output stream by the DACIS **103** of FIG. 1B, in accordance with an exemplary embodiment of the disclosure. Referring to the exemplary scenario **300**, there are illustrated two disparate live media output streams **302** and **304** published for two users associated with the client devices **132a** and **132n**, respectively. The disparate live media output stream **302** may be existing disparate live media output stream for the client device **132a**. The disparate live media output stream **304** may be a disparate per-client live media output stream published based on dynamic insertion of targeted non-programming content and customized programming content by the DACIS **103** of FIG. 1B for the client device **132n**.

The live NBA game may be scheduled to start at 8:00 pm. However, a user associated with the client device **132n** tunes in early to watch the NBA game, for example at 7:15 pm. However, the user may not be interested in watching the end of the current programming media content. In such a case, the PEM **115** in the DACIS **103** may determine targeted non-programming content, such as sports items ads, and customized programming content, such as NBA highlights of previous game and generates the alternate programming schedule **113**. Accordingly, the stream publishing engine **114** in the DACIS **103** may publish an alternate disparate live media output stream **117n** for the client device **132n**. At 8:00 pm, the DACIS **103** may switch back to the live NBA game scheduled at 8:00 pm.

FIGS. 4A to 4D depict flowcharts illustrating exemplary operations for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content in the media packaging and distribution system **102** of FIG. 1B, in accordance with an exemplary embodiment of the disclosure. Specifically, flowcharts **400A** and **400B** collectively depicts a method for just-in-time insertion of non-programming content and/or programming content, in accordance with an embodiment of the disclosure. Flowchart **400C** depicts a method for personalized insertion of non-programming content and/or programming content, in

accordance with an embodiment of the disclosure. Flowchart **400E** depicts a method for stream failover, in accordance with an embodiment of the disclosure.

Referring to FIGS. 4A and 4B, flowcharts **400A** and **400B** collectively depicts a method for just-in-time insertion of non-programming content and/or programming content, in accordance with an embodiment of the disclosure.

At **402A**, a first manifest request may be received from a first client device. In accordance with an embodiment, the DACIS **103** may be configured to receive the first manifest request from the first client device, such as a client device **132a**, to begin or continue playback of a first disparate live media output stream, for example the disparate live media output stream **117a**. The first manifest request may comprise one or more parameters. Examples of the one or more parameters may include, but are not limited to, universal resource locators and/or identifiers referencing records for existing one or more live input streams or one or more pre-encoded media assets in the CPDS **107**, a plurality of client-specific parameters, and a plurality of client-specified attributes derived from a user interaction with the first client device, such as the client device **132a**. Examples of the plurality of client-specific parameters may include, but are not limited to, user preferences and identifiers, client device preferences and identifiers, and one or more rules governed by geolocation data and current position of playback of a first disparate live media output stream, such as the disparate live media output stream **117a**, at the first client device, such as the client device **132a**. In an embodiment, the client-specified attributes derived from a user interaction with the first client device may include, but are not limited to, the user interaction with interactive content in a customized first programming content and a targeted first non-programming content, and a preference for a type and/or category of the targeted first non-programming content and/or the customized first programming content. The user interaction with the interactive content may comprise, for example, a selection to exclude the targeted first non-programming content and/or the customized first programming content, a selection to include a subset of the targeted first non-programming content and/or the customized first programming content, and a selection to include all of the targeted first non-programming content and/or the customized first programming content within one or more specified non-programming content locations.

At **404A**, the received first manifest request that comprises one or more parameters may be transmitted to the stream selection service **142**. In an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit the received first manifest request that comprises the one or more parameters to the stream selection service **142**.

In an embodiment, based on the received first manifest request, the stream selection service **142** may be configured to select one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**. Accordingly, the stream selection service **142** may be configured to return universal resource locators and/or identifiers that reference records for the selected one or more live input streams and/or one or more pre-encoded media assets to the PEM **115**.

At **406A**, the universal resource locators and/or identifiers that reference records for the selected one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user

preferences database **144** may be received. In an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the universal resource locators and/or identifiers that reference records for the selected one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**.

At **408A**, the received first manifest request may be transmitted to the CPDS **107** based on the selected one or more live input streams and/or one or more pre-encoded media assets. In an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit the received first manifest request to the CPDS **107** based on the selected one or more live input streams and/or one or more pre-encoded media assets.

In an embodiment, based on the received first manifest request, the CPDS **107** may be configured to retrieve indexed metadata, for example, non-programming indicators, such as ad break locations, graphical treatment indicators, such as overlay markers/triggers, programming indicators, such as SCTE35 markers and content duration. Accordingly, the CPDS **107** may be configured to return the indexed metadata to the PEM **115**.

At **410A**, the indexed metadata may be received from the CPDS **107**. In an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the indexed metadata from the CPDS **107**. Examples of the indexed metadata may include, but are not limited to, non-programming indicators, such as ad break locations, graphical treatment indicators, such as overlay markers/triggers, programming indicators, such as SCTE35 markers and content duration, as described above.

At **412A**, one or more content placement opportunities defined by the metadata associated with the one or more live input streams and/or one or more pre-encoded media assets may be identified. In an embodiment, the PEM **115** of the DACIS **103** may be configured to identify the one or more content placement opportunities defined by the metadata associated with the one or more live input streams and/or one or more pre-encoded media assets. Examples of the one or more content placement opportunities may include, but are not limited to, non-programming indicators, such as ad break locations, graphical treatment indicators, such as overlay markers/triggers, programming indicators, such as SCTE35 markers and personalized content opportunity. In an embodiment, for the identified one or more content placement opportunities, the PEM **115** of the DACIS **103** may be configured to determine additional content, such as non-programming content (such as advertisements), personalized programming content (such as promotional content), graphical treatment (such as overlays), and one or more decision point locations.

At **414**, a request may be transmitted to the Ad decisioning server **106a** to determine targeted first non-programming content. In an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit the request to the Ad decisioning server **106a** to determine the targeted first non-programming content.

In an embodiment, based on the request, the Ad decisioning server **106a** may be configured to identify the targeted first non-programming content, such as an advertising or an graphical treatment content, to be scheduled in one or more content placement opportunities, based on the one or more parameters. In an embodiment, the Ad decisioning server **106a** may be configured to identify the targeted first non-programming content based on execution of a non-program-

ming content service based on rules and conditions defined in additional parameters of the repository of schedules, rights, and user preferences database **144**, and the one or more parameters defined in the first manifest request. Accordingly, the Ad decisioning server **106a** may be configured to transmit the identified targeted first non-programming content to the PEM **115** of the DACIS **103**.

At **416**, the targeted first non-programming content may be received from the Ad decisioning server **106a**. In an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the targeted first non-programming content from the Ad decisioning server **106a**.

At **418**, a request may be transmitted to the CDS **105** to determine a customized first programming content to match one or more personalization parameters. In an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit the request to the CDS **105**. The CDS **105** may determine a customized first programming content to match one or more personalization parameters.

In an embodiment, based on the request, the CDS **105** may be configured to identify the customized first programming content based on the one or more parameters in the first manifest request, and the rules and conditions defined in the additional parameters of the repository of schedules, rights, and user preferences database **144**. Accordingly, the CDS **105** may be configured to transmit the identified customized first programming content to the PEM **115** of the DACIS **103**.

At **420**, the customized first programming content may be received from the CDS **105**. In an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the customized first programming content from the CDS **105**.

At **422**, a first additional content comprising the customized first programming content and the targeted first non-programming content may be determined for the first client device based on the one or more parameters and the associated indexed metadata. In an embodiment, the PEM **115** of the DACIS **103** may be configured to determine the first additional content comprising the customized first programming content, received from the CDS **105**, and the targeted first non-programming content, received from the Ad decisioning server **106a**. The first additional content may be determined for the first client device, such as the client device **132a**, based on the one or more parameters and the associated indexed metadata, received from the CPDS **107**.

At **424**, a first programming schedule **111** may be generated for the first client device based on the selected one or more live input streams and/or the one or more pre-encoded media assets, the associated indexed metadata, and the determined first additional content. In an embodiment, the PEM **115** of the DACIS **103** may be configured to generate the first programming schedule **111** for the first client device, such as the client device **132a**, based on selected one or more live input streams and/or the one or more pre-encoded media assets, the associated indexed metadata, and the determined first additional content.

In accordance with an embodiment, the generated first programming schedule **111** may not include the first additional content, in accordance with the one or more parameters in the first manifest request. In accordance with another embodiment, the generated first programming schedule **111** may include the first additional content, in accordance with the one or more parameters in the first manifest request. In accordance with another embodiment, the generated first programming schedule **111** may include a subset of the first additional content, in accordance with the one or more parameters in the first manifest request.

In accordance with an embodiment, the first additional content scheduled to be inserted, may be inserted into, for example, one or more pre-encoded assets in accordance with appropriate markers. The appropriate markers may be defined in the metadata of the one or more pre-encoded assets, and rules and conditions defined in the repository of schedule, rights, and user preferences database **144**.

In accordance with an embodiment, the generated first programming schedule **111** may define locations and types of one or more decision points defined by the one or more content placement opportunities in the indexed metadata. The one or more decision points may be included in the first programming schedule **111**. Within the first programming schedule **111**, the one or more decision points may define various types of the decision points, for example, an Ad preference, skip next ad, skip future ads, watch ads immediately instead of at future ad locations, or personalized content preferences.

In an exemplary embodiment, the one or more pre-encoded media assets are scheduled and the one or more decision points, defined by the one or more content placement opportunities, are inserted in the first programming schedule **111**. In such exemplary embodiment, the first programming schedule **111** may be configured to control the one or more pre-encoded media asset manifests to be published as, for example, the first disparate live media output stream manifest. Further, the first programming schedule **111** may include, for example, segments, markers, the one or more content placement opportunities, up to and including next available decision point.

At **426**, the first programming schedule **111** may be delivered to the stream publishing engine **114**. In accordance with an embodiment, the PEM **115** in the DACIS **103** may be configured to deliver the first programming schedule **111** to the stream publishing engine **114**.

At **428**, one or more live input stream manifests and/or one or more pre-encoded media asset manifests published in the content delivery system **130** may be selected. In accordance with an embodiment, the stream publishing engine **114** in the DACIS **103** may be configured to select the one or more live input stream manifests and/or the one or more pre-encoded media asset manifests published in the content delivery system **130**.

At **430**, a plurality of media segments indicated by the one or more pre-encoded media asset manifests and the one or more live input stream manifests, the associated indexed metadata, the targeted first non-programming content, and the customized first programming content may be indexed based on the first programming schedule **111** generated for the first client device. In accordance with an embodiment, the indexing and storage system **116** may be configured to index the plurality of media segments indicated by the one or more pre-encoded media asset manifests and the one or more live input stream manifests, the associated indexed metadata, the targeted first non-programming content, and the customized first programming content may be indexed based on the first programming schedule **111** generated for the first client device, for example the client device **132a**.

At **432**, a first disparate live media output stream manifest for the first client device may be published based on insertion of the selected one or more live input stream manifests and/or the one or more pre-encoded media asset manifests, the associated indexed metadata, and the determined first additional content, into the first disparate live media output stream manifest, in accordance with the first programming schedule **111** generated for the first client device. In accordance with an embodiment, the stream publishing engine

114 in the DACIS **103** may be configured to publish the first disparate live media output stream manifest for the first client device, for example the client device **132a**. The publication may be based on insertion of the selected one or more live input stream manifests and/or the one or more pre-encoded media asset manifests, the associated indexed metadata, and the determined first additional content, into the first disparate live media output stream manifest, in accordance with the first programming schedule **111** generated for the first client device, for example the client device **132a**.

At **434**, the published first disparate live media output stream manifest may be transmitted to a media player of the first client device for playback of the first disparate live media output stream, for example the disparate live media output stream **117a**. In accordance with an embodiment, the DACIS **103** may be configured to transmit the published first disparate live media output stream manifest to the media player of the first client device for playback of the first disparate live media output stream, for example the disparate live media output stream **117a**. In accordance with an embodiment, the first disparate live media output stream manifest for the first client device, for example the client device **132a**, may be generated based on one of a pre-defined conversion modes. The pre-defined conversion modes may correspond to conversion of pre-encoded media assets to live stream mode, pre-encoded media assets to live stream mode with scalable architecture, a live stream to live stream mode, and a mixed mode corresponding to switches between pre-encoded media assets and live streams.

In accordance with an embodiment, the published first disparate live media output stream manifest may be delivered to the first client device, for example the client device **132a**. Accordingly, the media player of the first client device, for example the client device **132a**, may begin or continue playback of the first disparate live media output stream **117a**. Further, the media player, during playback of the first disparate live media output stream **117a** at the first client device, for example the client device **132a**, presents one or more decision points defined by the interactive content to initiate a user interaction at the first client device, for example the client device **132a**.

In accordance with an embodiment, during the playback, in case a decision point exists, is enabled, and presented by the first client device, for example the client device **132a**, the user of the client device **132a** may interact with one or more interactive elements at the decision point and influence (or modify) remaining first disparate live media output stream **117a**.

For example, in an embodiment of just-in-time non-programming content insertion, at the one or more decision points, the user interaction corresponds to one of a desired selection corresponding to the user interaction with one or more interactive elements, or a default selection corresponding to non-interaction of a user with the customized first programming content and/or the targeted first non-programming content in the first disparate live media output stream **117a** played back by the media player. In another example, in an embodiment of personalized content playlist, at the one or more decision points, the user interaction corresponds to one of exclusion of the targeted first non-programming content or the customized first programming content and replacement by default content, selection of alternate customized first programming content, selection of a subsequent second programming content, approval or disapproval of the selected customized first programming content, exclusion of subsequent second non-programming content for a

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remaining portion of the first disparate live media output stream **117a** played back by the media player, selection of one or more targeted first non-programming content of a specific category, or viewing of some or all of the targeted first non-programming content immediately to avoid some or all of the targeted first non-programming content for the remaining portion of the first disparate live media output stream played back by the media player.

At **436**, the user interaction may be received that corresponds to one of a desired selection corresponding to the user interaction with one or more interactive elements, or a default selection corresponding to a non-interaction of the user with the customized first programming content and/or the targeted first non-programming content in the first disparate live media output stream **117a** played back by the media player. In accordance with an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the user interaction may be received. The user interaction may correspond to one of a desired selection corresponding to the user interaction with one or more interactive elements, or a default selection corresponding to a non-interaction of the user with the customized first programming content and/or the targeted first non-programming content in the first disparate live media output stream **117a** played back by the media player.

At **438**, a remaining portion of the first programming schedule **111** that generates the first disparate live media output stream **117a** corresponding to the first manifest request generated by the first client device, such as the client device **132a**, may be modified based on the user interaction with the interactive content. In accordance with an embodiment, the PEM **115** of the DACIS **103** may be configured to modify the remaining portion of the first programming schedule **111** that generates the first disparate live media output stream **117a** corresponding to the first manifest request generated by the first client device, such as the client device **132a**, based on the user interaction with the interactive content.

At **440**, a desired selection or the default selection may be transmitted to an external storage system that is the repository of schedules, rights, and user preferences database **144**. In accordance with an embodiment, the first client device, for example the client device **132a**, may be configured to transmit the desired selection or the default selection to the external storage system that is the repository of schedules, rights, and user preferences database **144**.

In accordance with an embodiment, the user interaction or non-interaction result may be sent to future client manifest requests to influence the remainder of the first disparate live media output stream **117a**. This may allow for a user choice, such as to skip an ad, to result with another manifest request to the DACIS **103** to update the existing first disparate live media output stream **117a** with the intended behavior and affect playback of the first disparate live media output stream **117a** within the first client device.

Referring to FIG. **4C**, flowchart **400C** depicts a method for personalized insertion of playlist of non-programming content and/or programming content, in accordance with an embodiment of the disclosure.

At **402B**, a first manifest request may be received from a first client device. In accordance with an embodiment, the DACIS **103** may be configured to receive the first manifest request from the first client device, such as a client device **132a**, to begin or continue playback of a first disparate live media output stream, for example the disparate live media output stream **117a**. The first manifest request may comprise one or more parameters. Examples of the one or more

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parameters may include, but are not limited to, universal resource locators and/or identifiers referencing records for existing one or more live input streams or one or more pre-encoded media assets in the CPDS **107**, a plurality of client-specific parameters, and a plurality of client-specified attributes derived from a user interaction with the first client device, such as the client device **132a**. Examples of the plurality of client-specific parameters may include, but are not limited to, user preferences and identifiers, client device preferences and identifiers, and one or more rules governed by geolocation data and current position of playback of a first disparate live media output stream, such as the disparate live media output stream **117a**, at the first client device, such as the client device **132a**. In another embodiment, the client-specified attributes derived from a user interaction with the first client device may include, but are not limited to, a preference for a type and/or category of the targeted first non-programming content and/or the customized first programming content, and a possible time constraint (or duration) to fill with the targeted first non-programming content and/or the customized first programming content. The user interaction with the interactive content may comprise, for example, a selection to exclude the targeted first non-programming content and/or the customized first programming content, a selection to include a subset of the targeted first non-programming content and/or the customized first programming content, and a selection to include all of the targeted first non-programming content and/or the customized first programming content within one or more specified non-programming content locations.

The user interaction with the interactive content may comprise, for example, a selection to exclude the targeted first non-programming content and/or the customized first programming content, a selection to include a subset of the targeted first non-programming content and/or the customized first programming content, and a selection to include all of the targeted first non-programming content and/or the customized first programming content within one or more specified non-programming content locations.

At **404B**, the received first manifest request that comprises one or more parameters may be transmitted to the stream selection service **142**. In an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit the received first manifest request that comprises the one or more parameters to the stream selection service **142**.

In an embodiment, based on the received first manifest request, the stream selection service **142** may be configured to select one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**. Accordingly, the stream selection service **142** may be configured to return universal resource locators and/or identifiers that reference records for the selected one or more live input streams and/or one or more pre-encoded media assets to the PEM **115**.

At **406B**, the universal resource locators and/or identifiers that reference records for the selected one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144** may be received. In an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the universal resource locators and/or identifiers that reference records for the selected one or more live input streams and/or one or more pre-encoded media assets based

on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**.

At **408B**, the received first manifest request may be transmitted to the CPDS **107** based on the selected one or more live input streams and/or one or more pre-encoded media assets. In an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit the received first manifest request to the CPDS **107** based on the selected one or more live input streams and/or one or more pre-encoded media assets.

In an embodiment, based on the received request, the CPDS **107** may be configured to retrieve indexed metadata, for example, non-programming indicators, such as ad break locations, graphical treatment indicators, such as overlay markers/triggers, programming indicators, such as SCTE35 markers, content duration, and categories (such as, "basketball", "sports", "Knicks") to which the programming content has been assigned.

At **410B**, the indexed metadata may be received from the CPDS **107**. In an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the indexed metadata from the CPDS **107**. Examples of the indexed metadata may include, but are not limited to, non-programming indicators, such as ad break locations, graphical treatment indicators, such as overlay markers/triggers, programming indicators, such as SCTE35 markers and content duration, as described above.

At **412B**, one or more content placement opportunities defined by the metadata associated with the one or more live input streams and/or one or more pre-encoded media assets may be identified. In an embodiment, the PEM **115** of the DACIS **103** may be configured to identify the one or more content placement opportunities defined by the metadata associated with the one or more live input streams and/or one or more pre-encoded media assets. Examples of the one or more content placement opportunities may include, but are not limited to, non-programming indicators, such as ad break locations, graphical treatment indicators, such as overlay markers/triggers, programming indicators, such as SCTE35 markers and personalized content opportunity. In an embodiment, for the identified one or more content placement opportunities, the PEM **115** of the DACIS **103** may be configured to determine additional content, such as non-programming content (such as advertisements), personalized programming content (such as promotional content), graphical treatment (such as overlays), and one or more decision point locations.

At **442**, a request may be transmitted to the CDS **105** to determine a customized first programming content from the CPDS **107** for matching one or more personalization parameters. In accordance with an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit a request to the CDS **105** to determine the customized first programming content from the CPDS **107** for matching one or more personalization parameters. Examples of the one or more personalization parameters may include, but are not limited to, user preferences and identifiers from the first manifest request or retrieved from the repository of schedules, rights, and user preferences database **144**, and time constraints. The time constraints may be determined based on user preferences retrieved from the repository of schedules, rights, and user preferences database **144**, a range defined in the first manifest request, and schedule tolerances defined in the repository of schedules, rights, and user preferences database **144**, client device preferences or identifiers from the first manifest request or retrieved from the repository of

schedules, rights, and user preferences database **144**, geo-location information from the first manifest request or retrieved from the repository of schedules, rights, and user preferences database **144**, and/or a content recommendation engine.

At **444**, the first programming schedule **111** may be generated for the first client device based on the selected one or more live input streams and/or the one or more pre-encoded media assets, the associated indexed metadata, and the determined first additional content. In accordance with an embodiment, the PEM **115** of the DACIS **103** may be configured to generate the first programming schedule **111** for the first client device, such as the client device **132a**, based on the selected one or more live input streams and/or the one or more pre-encoded media assets, the associated indexed metadata, and the determined first additional content.

In accordance with an embodiment, the first additional content scheduled to be inserted, may be inserted into, for example, one or more pre-encoded assets in accordance with appropriate markers. The appropriate markers may be defined in the metadata of the one or more pre-encoded assets, and rules and conditions defined in the repository of schedule, rights, and user preferences database **144**.

In accordance with an embodiment, the generated first programming schedule **111** may define locations and types of one or more decision points defined by the one or more content placement opportunities in the indexed metadata. Such one or more decision points may be included in the first programming schedule **111**. Within the first programming schedule **111**, the one or more decision points may define personalized content preferences, skipping of personalized content and moving to default content, and approval or disapproval of the personalized content.

In an exemplary embodiment, the one or more pre-encoded media assets are scheduled and the one or more decision points, defined by the one or more content placement opportunities, are inserted in the first programming schedule **111**. In such exemplary embodiment, the first programming schedule **111** may be configured to control the one or more pre-encoded media asset manifests to be published as, for example, the first disparate live media output stream manifest. Further, the first programming schedule **111** may include, for example, segments, markers, the one or more content placement opportunities, up to and including next available decision point. Thereafter, control passes to **426** in flowchart **400B** and exemplary operations till **440** may be performed in similar manner as described above.

Referring to FIG. **4D**, there is illustrated a flowchart **400D** depicting a method for stream failover, in accordance with an embodiment of the disclosure.

At **402C**, a first manifest request may be received from a first client device. In accordance with an embodiment, the DACIS **103** may be configured to receive the first manifest request from the first client device, such as a client device **132a**, to begin or continue playback of a first disparate live media output stream, for example the disparate live media output stream **117a**. The first manifest request may comprise one or more parameters. Examples of the one or more parameters may include, but are not limited to, universal resource locators of existing one or more live input streams, identifiers referencing records for existing one or more live input streams or one or more pre-encoded media assets in the CPDS **107**, and a plurality of client-specific parameters. Examples of the plurality of client-specific parameters may include, but are not limited to, user preferences and identi-

fiers, client device preferences and identifiers, and one or more rules governed by geolocation data and current position of playback of a first disparate live media output stream, such as the disparate live media output stream **117a**, at the first client device, such as the client device **132a**.

At **404C**, received first manifest request that comprises one or more parameters may be transmitted to the stream selection service **142**. In accordance with an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit the received first manifest request that comprises the one or more parameters to the stream selection service **142**.

At **454**, it may be determined that a set of criteria associated with the first disparate live media output stream is satisfied. In accordance with an embodiment, the stream selection service **142** may be configured to determine if the set of criteria associated with the first disparate live media output stream is satisfied. The set of criteria may include an accessibility of the first disparate live media output stream, update of the first disparate live media output stream and/or compatibility of the first disparate live media output stream having media and/or a manifest with the first manifest request. The stream selection service **142** may be further configured to determine the one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters in the first manifest request, and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**. The stream selection service **142** may be further configured to determine the one or more live input streams and/or one or more pre-encoded media assets based on rules provided by a stream owner/operator (such as a regional blackout for the first client device leading to an alternative stream to watch) and user preferences (that exclude certain categories) defined in the repository of schedules, rights, and user preferences database **144**.

In accordance with an embodiment, when the first disparate live media output stream is not accessible, the first disparate live media output stream is not updated and/or the first disparate live media output stream having media and/or manifest that is not compatible with the first manifest request. Further, in an absence of one or more live input streams, the stream selection service **142** may be configured to select a pre-encoded asset indicated in the first manifest request to continue playback as the first disparate live media output stream. In such embodiment, control passes to operation **456**. Otherwise, control passes to operation **406A** of the flowchart **400A**.

At **456**, universal resource locators and/or identifiers that reference records for the selected one or more live input streams or alternate one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144** may be received. In accordance with an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the universal resource locators and/or identifiers that reference records for the selected one or more live input streams or alternate one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**.

At **458**, a received first manifest request may be transmitted to the CPDS **107** based on the selected one or more live input streams or alternate one or more pre-encoded media assets. In accordance with an embodiment, the PEM **115** of the DACIS **103** may be configured to transmit the received first manifest request to the CPDS **107** based on the

selected one or more live input streams or alternate one or more pre-encoded media assets.

At **460**, alternate metadata for alternate additional content may be received from the CPDS **107**. In accordance with an embodiment, the PEM **115** of the DACIS **103** may be configured to receive the alternate metadata for the alternate additional content from the CPDS **107**. The metadata may include, for example, Ad break locations, overlay markers/triggers, SCTE35 markers/triggers, content duration, and one or more decision points. The alternate metadata may further include a location to transition from the first additional content to the alternate additional content.

At **462**, a placement of alternate additional content within the first disparate live media output stream manifest may be determined based on the associated indexed metadata and the alternate metadata. In accordance with an embodiment, the PEM **15** of the DACIS **103** may be configured to determine the placement of the alternate additional content within the first disparate live media output stream manifest based on the associated indexed metadata and the alternate metadata.

At **464**, an alternate programming schedule may be generated for the first client device based on the alternate additional content and alternate metadata. In accordance with an embodiment, the PEM **15** of the DACIS **103** may be configured to generate the alternate programming schedule for the first client device, such as the client device **132a**, based on the alternate additional content and alternate metadata.

At **466**, a final disparate live output stream manifest may be published for the first client device based on the generated alternate programming schedule. In accordance with an embodiment, the PEM **15** of the DACIS **103** may be configured to publish the final disparate live output stream manifest for the first client device, such as the client device **132a**, based on the generated alternate programming schedule.

In accordance with an embodiment, the final disparate live output stream manifest for the first client device may be generated based on one of the pre-defined conversion modes. The pre-defined conversion modes may correspond to pre-encoded media assets to live stream mode, pre-encoded media assets to live stream mode with scalable architecture, a live stream to live stream mode, and a mixed mode corresponding to switches between pre-encoded media assets and live streams.

In accordance with an embodiment, during ployout, the first additional content may be transitioned to the alternate additional content based on one or more transition parameters. The one or more transition parameters may comprise one or more parameters from the first manifest request, current state of the first disparate live media output stream manifest determined based on accessibility, regular update, and suitable encoding, digital rights management, and compatibility with the first client device, rules provided by a stream owner operator, and user preferences defined in the repository of schedules, rights, and user preferences database **144**.

As described above in FIGS. **4A** and **4B**, the published final disparate live output stream manifest may be transmitted to the media player of the first client device for playback of the final disparate live media output stream. Accordingly, the first client device begins or continues the playback of the final disparate live media output stream.

In an alternate embodiment, at **454**, the stream selection service **142** may be configured to select a second disparate live media output stream according to accessibility of the

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first disparate live media output stream in case the set of criteria associated with the first disparate live media output stream is not satisfied. The stream selection service **142** may be configured to select the second disparate live media output stream according to rules provided by a stream owner/operator and user preferences defined in the repository of schedules, rights, and user preferences database **144**. The operations that follow may be performed in a similar manner, as the operations for the first disparate live media output stream are performed.

FIG. **5** is a conceptual diagram illustrating an example of a hardware implementation for a media packaging and distribution system **102** employing a processing system for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content, in accordance with an exemplary embodiment of the disclosure. Referring to FIG. **5**, the hardware implementation shown by a representation **500** for the media packaging and distribution system **102** employs a processing system **502** for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content, in accordance with an exemplary embodiment of the disclosure, as described herein.

In some examples, the processing system **502** may comprise one or more hardware processors **504**, a non-transitory computer-readable medium **506**, a bus **508**, a bus interface **510**, and a transceiver **512**. FIG. **5** further illustrates the DACIS **103**, the CDS **105**, the CPDS **107**, the stream publishing engine **114**, PEM **115**, indexing and storage system **116**, the stream selection service **142**, and the repository of schedules, rights, and user preferences database **144**, as described in detail in FIGS. **1A** and **1B**.

The hardware processor **504** may be configured to manage the bus **508** and general processing, including the execution of a set of instructions stored on the non-transitory computer-readable medium **506**. The set of instructions, when executed by the processor **504**, causes the media packaging and distribution system **102** to execute the various functions described herein for any particular apparatus. The hardware processor **504** may be implemented, based on a number of processor technologies known in the art. Examples of the hardware processor **504** may be a Reduced Instruction Set Computing (RISC) processor, an Application-Specific Integrated Circuit (ASIC) processor, a Complex Instruction Set Computing (CISC) processor, and/or other processors or control circuits.

The non-transitory computer-readable medium **506** may be used for storing data that is manipulated by the processor **504** when executing the set of instructions. The data is stored for short periods or in the presence of power. The non-transitory computer-readable medium **506** may also be configured to store data for one or more of the DACIS **103**, the CDS **105**, the CPDS **107**, the stream publishing engine **114**, PEM **115**, indexing and storage system **116**, the stream selection service **142**, and the repository of schedules, rights, and user preferences database **144**.

The bus **508** is configured to link together various circuits. In this example, the media packaging and distribution system **102** employing the processing system **502** and the non-transitory computer-readable medium **506** may be implemented with bus architecture, represented generally by bus **508**. The bus **508** may include any number of interconnecting buses and bridges depending on the specific implementation of the media packaging and distribution system **102** and the overall design constraints. The bus interface **510**

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may be configured to provide an interface between the bus **508** and other circuits, such as, transceiver **512**, and external devices, such as source device **118**, external data source **120**, and client devices **132a**, . . . , **132n**.

The transceiver **512** may be configured to provide a communication of the media packaging and distribution system **102** with various other apparatus, such as the Ad decisioning servers **106a**, . . . , **106n**, the consumer devices **110a**, . . . , **110n**, such as the client devices **132a**, . . . , **132n**, the external data source **120**, and the source device **118**, via the network **108**. The transceiver **512** may communicate via wireless communication with networks, such as the Internet, the Intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (WLAN) and/or a metropolitan area network (MAN). The wireless communication may use any of a plurality of communication standards, protocols and technologies, such as Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), Long Term Evolution (LTE), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Wireless Fidelity (Wi-Fi) (such as IEEE 802.11a, IEEE 802.11b, IEEE 802.11g and/or IEEE 802.11n), voice over Internet Protocol (VoIP), and/or Wi-MAX.

It should be recognized that, in some embodiments of the disclosure, one or more components of FIG. **5** may include software whose corresponding code may be executed by at least one processor, for across multiple processing environments. For example, the DACIS **103**, the CDS **105**, the CPDS **107**, the stream publishing engine **114**, PEM **115**, indexing and storage system **116**, the stream selection service **142**, and the repository of schedules, rights, and user preferences database **144** may include software that may be executed across a single or multiple processing environments.

In an aspect of the disclosure, the processor **504**, the non-transitory computer-readable medium **506**, or a combination of both may be configured or otherwise specially programmed to execute the operations or functionality of the DACIS **103**, the CDS **105**, the CPDS **107**, the stream publishing engine **114**, PEM **115**, indexing and storage system **116**, the stream selection service **142**, and the repository of schedules, rights, and user preferences database **144**, or various other components described herein, as described with respect to FIGS. **1A** to **4D**.

Various embodiments of the disclosure comprise the media packaging and distribution system **102** that may be configured to publish disparate live media output streams to be viewed on a plurality of consumer devices (such as the consumer devices **110a**, . . . , **110n**) based on user selection. The media packaging and distribution system **102** may comprise, for example, the DACIS **103**, the CDS **105**, the CPDS **107**, the stream publishing engine **114**, PEM **115**, indexing and storage system **116**, the stream selection service **142**, and the repository of schedules, rights, and user preferences database **144**. In accordance with an embodiment, one or more processors in the DACIS **103** may be configured to receive a first manifest request, comprising one or more parameters, from a first client device, such as client device **132a**. The one or more processors in the DACIS **103** may be further configured to determine a first additional content comprising a customized first programming content and a targeted first non-programming content for the first client device based on the one or more parameters and associated indexed metadata retrieved based on the one or more parameters. The one or more processors in the

DACIS 103 may be further configured to generate the first programming schedule 111, for the first client device based on selected one or more live input stream and/or the one or more pre-encoded media assets, the associated indexed metadata, and the determined first additional content. The one or more processors in the DACIS 103 may be further configured to select one or more live input stream manifests and/or one or more pre-encoded media asset manifests published in the content delivery system 130 and associated indexed metadata based on the one or more parameters. The one or more processors in the DACIS 103 may be further configured to publish a first disparate live media output stream manifest for the first client device based on insertion of the selected one or more live input stream manifests and/or the one or more pre-encoded media asset manifests, the associated indexed metadata, and the determined first additional content in accordance with the first programming schedule 111, generated for the first client device.

In accordance with an embodiment, the one or more processors in the DACIS 103 may be further configured to index a plurality of media segments indicated by the one or more pre-encoded media asset manifests and the one or more live input stream manifests, the associated indexed metadata, the targeted first non-programming content, and the customized first programming content based on the first programming schedule 111 generated for the first client. In accordance with an embodiment, the one or more processors in the DACIS 103 may be further configured to insert the first additional content at one or more content placement opportunities indicated in the first disparate live media output stream manifest based on the first programming schedule 111 generated for the first client device. The one or more content placement opportunities may be defined by the metadata associated with the one or more live input streams corresponding to the one or more live input stream manifests and/or one or more pre-encoded media assets corresponding to the one or more pre-encoded media asset manifests. The insertion of the targeted first non-programming content and the customized first programming content in real time, the associated indexed metadata and/or the one or more content placement opportunities may include programming content indicators, non-programming content indicators, graphical treatment indicators, and interactive content indicators.

In accordance with an embodiment, the insertion of the customized first programming content, in an instance in which the generated first disparate live media output stream manifest corresponds to a disparate live media output stream, a first set of processors in the DACIS 103 may be further configured to receive universal resource locators and/or identifiers referencing records for the selected one or more live input streams and/or the one or more pre-encoded media assets from the CPDS 107. The associated indexed metadata and/or one or more content placement opportunities may include programming content indicators, non-programming content indicators, graphical treatment indicators, and interactive content indicators. The programming content indicators may comprise a plurality of categories for the customized first programming content.

In accordance with an embodiment, the first set of processors in the DACIS 103 may be further configured to transmit a request to the CDS 105 for determining the customized first programming content from a content package and distribution system for matching one or more personalization parameters. The one or more personalization parameters may include user preferences and identifiers from the first manifest request or retrieved from the repository of schedules, rights, and user preferences database 144,

and time constraints. The time constraints may be determined based on user preferences retrieved from the repository of schedules, rights, and user preferences database 144, a range defined in the first manifest request, and schedule tolerances defined in the repository of schedules, rights, and user preferences database 144, client device preferences or identifiers from the first manifest request or retrieved from the repository of schedules, rights, and user preferences database 144, geolocation information from the first manifest request or retrieved from the repository of schedules, rights, and user preferences database 144, and/or a content recommendation engine.

In accordance with an embodiment, the one or more parameters may comprise universal resource locators and/or identifiers referencing records for one or more live input streams or one or more pre-encoded media assets in the CPDS 107, a plurality of client-specific parameters, and a plurality of client-specified attributes derived from a user interaction with the first client device. The plurality of client-specific parameters may comprise user preferences and identifiers, client device preferences and identifiers, and one or more rules governed by geolocation data and current position of playback of a first disparate live media output stream at the first client device. In accordance with an embodiment, for the insertion of the targeted first non-programming content, the plurality of client-specified attributes may comprise the user interaction with interactive content in the customized first programming content and the targeted first non-programming content, and a preference for a type and/or category of the targeted first non-programming content and/or the customized first programming content. The user interaction with the interactive content comprises a selection to exclude the targeted first non-programming content and/or the customized first programming content, a selection to include a subset of the targeted first non-programming content and/or the customized first programming content, and a selection to include all of the targeted first non-programming content and/or the customized first programming content within one or more specified non-programming content locations. The targeted first non-programming content may comprise personalized advertisements including video advertisements, graphical treatment, cue points, and the interactive content comprising a set of interactive elements for the targeted first non-programming content. The customized first programming content may comprise personalized non-advertising content including promotional content, a short-form content, and an alternate additional content for replacement of at least a portion of a first disparate live media output stream generated for the first client.

In accordance with an embodiment, the one or more processors in the DACIS 103 may be further configured to transmit the published first disparate live media output stream manifest to a media player of the first client device for a playback of a first disparate live media output stream. The media player, during playback of the first disparate live media output stream at the first client device, may present one or more decision points defined by the interactive content to initiate a user interaction at the first client device. At the one or more decision points, the user interaction may correspond to one of a desired selection corresponding to the user interaction with one or more interactive elements, or a default selection corresponding to non-interaction of a user with the customized first programming content and/or the targeted first non-programming content in the first disparate live media output stream played back by the media player. The desired selection or the default selection may be trans-

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mitted to an external storage system that is the repository of schedules, rights, and user preferences database **144**. At the one or more decision points, the one or more interactive elements may be configured to exclude the targeted first non-programming content or the customized first programming content and be replaced by default content, select alternate customized first programming content, select a subsequent second programming content, approve or disapprove selected customized first programming content, exclude subsequent second non-programming content for a remaining portion of the first disparate live media output stream played back by the media player, select one or more targeted first non-programming content of a specific category, or view some or all of the targeted first non-programming content immediately to avoid some or all of the targeted first non-programming content for the remaining portion of the first disparate live media output stream played back by the media player.

In accordance with an embodiment, the one or more processors in the DACIS **103** may be further configured to modify a remaining portion of the first programming schedule **111** that generates first disparate live media output stream corresponding to the first manifest request generated by the first client device based on the user interaction with the interactive content. A first set of processors in the DACIS **103** may be configured to receive the first manifest request from a second set of processors in the DACIS **103**, select, based on the received first manifest request, one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**, and transmit universal resource locators and/or identifiers referencing records for the selected one or more live input streams and/or the one or more pre-encoded media assets to the second set of processors in the DACIS **103**.

In accordance with an embodiment, the second set of processors in the DACIS **103** may be further configured to retrieve the indexed metadata associated with the selected one or more live input streams and/or the one or more pre-encoded media assets from the CPDS **107**. The associated indexed metadata may define one or more content placement opportunities within the selected one or more live input streams and/or the one or more pre-encoded media assets. The customized first programming content and the targeted first non-programming content may be identified as the first additional content to be scheduled at the one or more content placement opportunities based on the one or more parameters in the received first manifest request and the associated indexed metadata, wherein the generated first programming schedule **111** excludes the first additional content or includes the first additional content completely or partially.

In accordance with an embodiment, a third set of processors in the DACIS **103** may be further configured to insert, based on the generated first programming schedule **111**, the identified customized first programming content and the targeted first non-programming content into the first disparate live media output stream manifest in accordance with the one or more content placement opportunities defined in the associated indexed metadata and one or more rules and conditions defined in the repository of schedules, rights, and user preferences database **144**. The identification of the targeted first non-programming content may be based on execution of a non-programming content service based on rules and conditions defined in additional parameters of the repository of schedules, rights, and user preferences data-

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base **144**, and the one or more parameters defined in the first manifest request. The identification of the customized first programming content may be based on execution of the CDS **105** based on the rules and conditions defined in the additional parameters of the repository of schedules, rights, and user preferences database **144**. In accordance with an embodiment, the first programming schedule **111** may define locations and types of one or more decision points defined by one or more content placement opportunities in the indexed metadata. In an instance when one or more pre-encoded media assets are being scheduled and the one or more decision points defined by the one or more content placement opportunities are to be inserted in the first programming schedule **111**, the first programming schedule **111** may be configured to control the one or more pre-encoded media asset manifests to be published as the first disparate live media output stream manifest.

In accordance with an embodiment, a conversion of one or more pre-encoded media assets into a first disparate live media output stream facilitates one or more subsequent modifications on the first disparate live media output stream. The one or more subsequent modifications may correspond to a user selection, a user preference, a change in the first programming schedule **111**, or a time or geolocation-based rule.

In accordance with an embodiment, the first set of processors in the DACIS **103** may be further configured to transmit universal resource locators and/or identifiers referencing records for the selected one or more live input streams and/or the one or more pre-encoded media assets to the second set of processors. In an absence of one or more live input streams, the first set of processors may be configured to select a pre-encoded asset indicated in the first manifest request to continue playback as the first disparate live media output stream. The second set of processors may be configured to select a second disparate live media output stream according to accessibility of the first disparate live media output stream. The first disparate live media output stream is not updated and/or the first disparate live media output stream has media and/or a manifest that is incompatible with a client request. The second set of processors may be further configured to select the second disparate live media output stream according to rules provided by a stream owner/operator and user preferences defined in the repository of schedules, rights, and user preferences database **144**.

In accordance with an embodiment, the one or more processors in the DACIS **103** may be further configured to transition from the first additional content to an alternate additional content based on one or more transition parameters. The one or more transition parameters may comprise one or more parameters from the first manifest request, current state of the first disparate live media output stream manifest determined based on accessibility, regular update, and suitable encoding, digital rights management, and compatibility with the first client device, rules provided by a stream owner operator, and user preferences defined in the repository of schedules, rights, and user preferences database **144**.

In accordance with an embodiment, the one or more processors in the DACIS **103** may be further configured to retrieve alternate metadata for the alternate additional content from the CPDS **107**. The alternate metadata may indicate a location to transition from the first additional content to the alternate additional content. The one or more processors may be further configured to determine placement of the alternate additional content within the first disparate live media output stream manifest based on the

associated indexed metadata and the alternate metadata. The one or more processors may be further configured to generate the alternate programming schedule **113**, for the first client device based on the alternate additional content and the alternate metadata, and transmit the generated alternate programming schedule to the first set of processors. The first set of processors may be configured to generate a final disparate live output stream manifest for the first client device. In accordance with an embodiment, another conversion of the first disparate live media output stream into the one or more pre-encoded media assets facilitates download of the one or more pre-encoded media assets at the first client device and mitigates dependency on the system for playback of remaining portion. In accordance with an embodiment, the one or more processors in the DACIS **103** may be further configured to receive a request from the first client device, wherein the request comprises the one or more parameters. The one or more processors in the DACIS **103** may be further configured to determine alternate additional content for inclusion in the first disparate live media output stream manifest or replacement of the first additional content in the first disparate live media output stream manifest based on the one or more parameters. The one or more processors in the DACIS **103** may be further configured to modify the first programming schedule **111** to generate an alternate programming schedule for the first client device based on the alternate additional content and associated metadata. The one or more processors in the DACIS **103** may be further configured to transmit the generated alternate programming schedule to a second set of processors, wherein the second set of processors is configured to generate a final disparate live output stream manifest for the first client device.

In accordance with an embodiment, the one or more processors in the DACIS **103** may be further configured to revoke the first disparate live media output stream manifest published for the first client device based on an identifier primitive associated with the first disparate live media output stream of the first client device in an instance in which a media player of the first client device is determined to be a plagiarized media player. The first disparate live media output stream may include at least one unique identifier inserted by the one or more processors in the DACIS **103**.

In accordance with various embodiments, the first disparate live media output stream manifest for the first client device may be generated based on one of a pre-defined conversion modes. The pre-defined conversion modes may correspond to pre-encoded media assets to live stream mode, pre-encoded media assets to live stream mode with scalable architecture, a live stream to live stream mode, and a mixed mode corresponding to switching between pre-encoded media assets and live streams.

In accordance with an embodiment, one or more processors in the PEM **115** may be configured to receive a first manifest request, comprising one or more parameters, from the first client device. Further, the received first manifest request may be transmitted to a first set of processors in the stream selection service **142**. The first set of processors may be configured to select, based on the received first manifest request, one or more live input streams and/or one or more pre-encoded media assets based on the one or more parameters and additional parameters retrieved from the repository of schedules, rights, and user preferences database **144**. The one or more processors in the PEM **115** may be configured to receive universal resource locators and/or identifiers referencing records for the selected one or more live input streams and/or the one or more pre-encoded media assets

from the first set of processors. The one or more processors in the PEM **115** may be configured to retrieve indexed metadata associated with the selected one or more live input streams and/or the one or more pre-encoded media assets from a content packaging and distribution system, wherein the indexed metadata defines one or more content placement opportunities within the selected one or more live input streams and/or the one or more pre-encoded media assets. The one or more processors in the PEM **115** may be further configured to identify a customized first programming content and a targeted first non-programming content as a first additional content to be scheduled in the one or more content placement opportunities based on the one or more parameters in the received first manifest and the associated indexed metadata. The one or more processors in the PEM **115** may be further configured to generate the first programming schedule **111** for the first client device based on the associated indexed metadata, the first additional content and the one or more live input streams and/or one or more pre-encoded media assets.

Various embodiments of the disclosure may provide a computer-readable medium, such as the non-transitory computer-readable medium **506**, having stored thereon, computer implemented instruction that when executed by the processor **504** causes the media packaging and distribution system **102** to execute operations for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content. In accordance with an embodiment, the processor **504** causes the media packaging and distribution system **102** to execute operations to receive a first manifest request, comprising one or more parameters, from a first client device, such as client device **132a**. The processor **504** causes the media packaging and distribution system **102** to execute operations to determine a first additional content comprising a customized first programming content and a targeted first non-programming content for the first client device based on the one or more parameters and associated indexed metadata retrieved based on the one or more parameters. The processor **504** causes the media packaging and distribution system **102** to execute operations to generate the first programming schedule **111**, for the first client device based on selected one or more live input stream and/or the one or more pre-encoded media assets, the associated indexed metadata, and the determined first additional content. The processor **504** causes the media packaging and distribution system **102** to execute operations to select one or more live input stream manifests and/or one or more pre-encoded media asset manifests published in the content delivery system **130** and associated indexed metadata based on the one or more parameters. The processor **504** causes the media packaging and distribution system **102** to execute operations to publish a first disparate live media output stream manifest for the first client device based on insertion of the selected one or more live input stream manifests and/or the one or more pre-encoded media asset manifests, the associated indexed metadata, and the determined first additional content in accordance with the first programming schedule **111**, generated for the first client device.

Existing systems for SSAI support live streaming and make decisions to insert non-programming content in near real-time. However, for On-Demand streaming, such decisions to insert non-programming content are made upfront. Accordingly, amount of control the user can excerpt to influence what non-programming content is shown may get limited. Further, newer ad models are developing that rely

less on interruptive video advertising and more on contextual based graphical treatment advertising within the program content. To this extent, SSAI systems have been able to circumvent ad blockers by requesting video ads on behalf of the client device and stitching them into the disparate live media output stream manifest.

To address at least the above problems, in accordance with the various embodiments of the present disclosure, the DACIS **103** in the media packaging and distribution system **102** may be configured to include not showing non-programming content that the user elected to skip or rated poorly in an earlier non-programming content break, or could enable the user to skip all non-programming content because the user interacted with an previous non-programming content or made a purchase and the advertiser elected to sponsor the remainder of the program.

The DACIS **103** may notified about graphical treatment opportunities within the content, make the necessary ad calls on behalf of the client device, and provide the client device with the information needed to execute the overlays via a secure out-of-band channel between the DACIS **103** and the client device. In many cases, a channel already exists to support passing the program indicators, such as start and end information, to the client device and Video Player Ad-Serving Interface (VPAID) ads that cannot be stitched.

DACIS **103** may also provide an opportunity to leverage the 1:1 scale of traditional SSAI systems to support custom content choices and not just targeted advertising. For example, when a user selects to join a live stream, the CDS **105** coupled with the DACIS **103** may determine that, instead of joining the live stream for the last few minutes of a program, the user should instead be shown content more relevant to the next program. For example, showing a personalized set of basketball highlights and ads to a user who likely joined the stream to watch the basketball game that is coming on next.

Further, the DACIS **103** may be used to provide seamless failover between redundant streams for large events, thus improving reliability. While some client devices support primary and backup streams and are able to fail between them, many client devices do not. In such cases, the client device may attempt to join the alternative stream after occurrence of an event, such as a device failure or crash. For such clients, the DACIS **103** monitors both the primary and backup stream, and if there is a failure, inserts the alternative stream into the output manifest. Thus, the media packaging and distribution system **102** provides an enhanced, intelligent, and personalized viewer experience with increased appeal in order to retain and gain a wider audience.

The DACIS **103**, the CDS **105**, the CPDS **107**, the stream publishing engine **114**, PEM **115**, indexing and storage system **116**, the stream selection service **142**, and the repository of schedules, rights, and user preferences database **144** in conjunction with each other, provide significant productivity and efficiency improvements since the process of generating disparate live media output streams with additional content is specific to each consumer device. The disparate live media output streams are simplified as the generated disparate live media output streams are independent of a requirement to re-process, that is re-encode and re-package, various live input streams for media distribution to the plurality of consumer devices in real time or near-real time. Thus, the network provider now may provide live channel offerings in a cost-effective manner.

Various components, as described above in FIG. 1B, enable the media packaging and distribution system **102** to leverage the modern streaming protocols, based on which

the media packaging and distribution system **102** generates disparate live media output streams of the live broadcast channels with additional content and one or more decision points to influence the remaining disparate live media output streams. The disparate live media output streams may be generated based on insertion of live input streams into a generated disparate live media output stream using pre-encoded media assets. In this regard, manifests from the live input streams and the pre-encoded media assets may be manipulated and produced for distribution of the disparate live media output stream.

As the media content itself does not need to be processed beyond the initial creation of the live input streams and pre-encoded media assets prepared for distribution, it is extremely inexpensive to provide such disparate live media output streams and alternate disparate live media output stream (in case of stream failover). It may be based solely on the cost to manipulate the manifests, which provide the instructions for the media players in the client devices **132a**, . . . , **132n** to execute. The media packaging and distribution system **102** may also support targeted ad insertion and customized programming content insertion on a per client basis and may further leverage the processing power of the individual client devices **132a**, . . . , **132n** to insert targeted channel graphics and graphical treatment advertisements and promotions.

As utilized herein the terms “circuits” and “circuitry” refer to physical electronic components (for example, hardware) and any software and/or firmware (“code”) which may configure the hardware, be executed by the hardware, and/or otherwise be associated with the hardware. As used herein, for example, a particular processor and memory may comprise a first “circuit” when executing first one or more lines of code and may comprise a second “circuit” when executing second one or more lines of code. As utilized herein, “and/or” means any one or more of the items in the list joined by “and/or”. As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. As utilized herein, the term “exemplary” means serving as a non-limiting example, instance, or illustration. As utilized herein, the terms “e.g.” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations. As utilized herein, circuitry is “operable” to perform a function whenever the circuitry comprises the necessary hardware and/or code (if any is necessary) to perform the function, regardless of whether performance of the function is disabled, or not enabled, by some user-configurable setting.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of embodiments of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”, “comprising”, “includes” and/or “including”, when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Further, many embodiments are described in terms of sequences of actions to be performed by, for example, elements of a computing device. It will be recognized that various actions described herein can be performed by specific circuits (e.g., application specific integrated circuits

(ASICs)), by program instructions being executed by one or more processors, or by a combination of both. Additionally, these sequences of actions described herein can be considered to be embodied entirely within any non-transitory form of computer readable storage medium having stored therein a corresponding set of computer instructions that upon execution would cause an associated processor to perform the functionality described herein. Thus, the various aspects of the disclosure may be embodied in a number of different forms, which have been contemplated to be within the scope of the claimed subject matter. In addition, for each of the embodiments described herein, the corresponding form of any such embodiments may be described herein as, for example, "logic configured to" perform the described action.

Another embodiment of the disclosure may provide a non-transitory machine and/or computer readable storage and/or media, having stored thereon, a machine code and/or a computer program having at least one code section executable by a machine and/or a computer, thereby causing the machine and/or computer to perform the steps as described herein for publishing a disparate per-client live media output stream based on dynamic insertion of targeted non-programming content and customized programming content.

The present disclosure may also be embedded in a computer program product, which comprises all the features enabling the implementation of the methods described herein, and which when loaded in a computer system is able to carry out these methods. Computer program in the present context means any expression, in any language, code or notation, either statically or dynamically defined, of a set of instructions intended to cause a system having an information processing capability to perform a particular function either directly or after either or both of the following: a) conversion to another language, code or notation; b) reproduction in a different material form.

Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, algorithms, and/or steps described in connection with the embodiments disclosed herein may be implemented as electronic hardware, computer software, firmware, or combinations thereof. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

The methods, sequences and/or algorithms described in connection with the embodiments disclosed herein may be embodied directly in firmware, hardware, in a software module executed by a processor, or in a combination thereof. A software module may reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, physical and/or virtual disk, a removable disk, a CD-ROM, virtualized system or device such as a virtual servers or container, or any other form of storage medium known in the art. An exemplary storage medium is communicatively coupled to the processor (including logic/code executing in the processor) such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor.

While the present disclosure has been described with reference to certain embodiments, it will be noted understood by, for example, those skilled in the art that various changes and modifications could be made and equivalents may be substituted without departing from the scope of the present disclosure as defined, for example, in the appended claims. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. The functions, steps and/or actions of the method claims in accordance with the embodiments of the disclosure described herein need not be performed in any particular order. Furthermore, although elements of the disclosure may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed, but that the present disclosure will include all embodiments falling within the scope of the appended claims.

What is claimed is:

1. A system, comprising at least one memory for storing instructions and one or more processors configured to:
 - generate a first programming schedule comprising one or more decision points defined by one or more content placement opportunities in an indexed metadata;
 - control publishing of one or more manifests based on a schedule of one or more pre-encoded media assets and the one or more decision points in the first programming schedule, wherein the one or more decision points indicate real-time facilitation relative to a playback of a media stream that is streamed on a media player of a first client device, and is associated with the one or more manifests reaching the one or more decision points, of one or more user interactions relating to one or more of content exclusion or content selection;
 - receive a first manifest request from the first client device, wherein the first manifest request is associated with at least one of the one or more manifests, and comprises one or more parameters, and wherein the one or more parameters comprise an indicator of one or more pre-encoded media assets; and
 - responsive to receiving the first manifest request:
 - transmit, to a second processor, an identifier representing one or more records for at least one of one or more live input streams and the one or more pre-encoded media assets;
 - in an absence of the one or more live input streams, select a pre-encoded media asset from the one or more pre-encoded media assets, based on the indicator of the pre-encoded media asset, in place of the one or more live inputs streams;
 - generate an updated manifest comprising the indicator of the pre-encoded media asset; and
 - publish the updated manifest to the second processor for streaming the pre-encoded media asset in place of the one or more live input streams for streaming the pre-encoded media asset via the media player of the first client device.
2. The system according to claim 1, wherein the one or more manifests are selected based on the one or more parameters, and wherein the one or more processors are further configured to:
 - determine first additional content comprising a customized first programming content and a targeted first non-programming content for the first client device based on the one or more parameters.

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3. The system according to claim 2, wherein the one or more processors are further configured to:

insert the first additional content at the one or more content placement opportunities based on the first programming schedule, wherein the one or more content placement opportunities comprise at least one of programming content indicators, non-programming content indicators, graphical treatment indicators, or interactive content indicators.

4. The system according to claim 3, wherein the one or more content placement opportunities comprise programming content indicators comprising a plurality of categories for the customized first programming content.

5. The system according to claim 4, wherein determining the customized first programming content comprises transmitting a request comprising the one or more parameters to a content decisioning service.

6. The system according to claim 3, wherein, the insertion of the first additional content is further based on the one or more user interactions, and the one or more user interactions indicate a preference for at least a type or a category of at least one of the targeted first non-programming content or the customized first programming content.

7. The system according to claim 2, wherein the one or more parameters comprise at least one of universal resource locators or identifiers, a plurality of client-specific parameters, and a plurality of client-specified attributes derived from one or more user interactions with the first client device, and wherein the plurality of client-specific parameters comprises user preferences and identifiers, client device preferences and identifiers, and one or more rules governed by geolocation data and position of playback of the media stream.

8. The system according to claim 2, wherein the targeted first non-programming content comprises personalized advertisements including video advertisements, graphical treatment, cue points, and an interactive content that comprises one or more interactive elements for the targeted first non-programming content, and

wherein the customized first programming content comprises personalized non-advertising content including promotional content, a short-form content, and an alternate additional content for replacement of at least a portion of the media stream.

9. The system according to claim 2, wherein the one or more processors are further configured to:

modify a portion of the first programming schedule based on the one or more user interactions in response to receiving the first manifest request.

10. The system according to claim 8, wherein the media stream comprises a default selection of content in response to one or more detections of non-interaction.

11. The system according to claim 10, wherein the one or more interactive elements indicate options for replacing the targeted first non-programming content or the customized first programming content with default content, selecting alternate customized first programming content, selecting a subsequent second programming content, approving selected customized first programming content, disapproving selected customized first programming content, excluding subsequent second non-programming content for a remaining portion of the media stream played back, selecting one or more targeted first non-programming content of a specific category, or viewing the targeted first non-programming content immediately responsive to one or more of

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the one or more user interactions indicating to avoid the targeted first non-programming content for a remaining portion of the media stream.

12. The system according to claim 2, wherein the one or more processors are further configured to:

transmit at least one of universal resource locators or identifiers that reference the one or more pre-encoded media assets.

13. The system according to claim 12, wherein the one or more processors are further configured to:

retrieve the indexed metadata from a content packaging and distribution system.

14. The system according to claim 2, wherein one or more processors are configured to:

insert, based on the first programming schedule, the customized first programming content and the targeted first non-programming content into the one or more manifests based on the one or more content placement opportunities in the indexed metadata and one or more rules and conditions defined in a repository of schedules, rights, and user preferences database.

15. The system according to claim 14, wherein the targeted first non-programming content is identified based on execution of a non-programming content service based on the one or more rules and conditions defined in additional parameters of the repository of schedules, rights, and user preferences database, and the one or more parameters of the first manifest request, wherein the customized first programming content is identified based on the one or more rules and conditions defined in the additional parameters of the repository of schedules, rights, and user preferences database.

16. The system according to claim 1, wherein the one or more processors are further configured to:

convert the one or more pre-encoded media assets into the media stream; and

facilitate one or more modifications of the media stream based on at least one of a user selection, a user preference, a change in the first programming schedule, or a time-based rule or geolocation-based rule.

17. The system according to claim 16, wherein the one or more processors are further configured to:

select a second media stream based on at least one of accessibility of the media stream, rules provided by a stream owner or a stream operator, and the user preference, wherein the media stream is unchanged or the media stream has at least one of media or a manifest that is incompatible with a client request.

18. The system according to claim 2, wherein the one or more processors are further configured to:

transition from the first additional content to an alternate additional content based on one or more transition parameters that comprise (a) the one or more parameters from the first manifest request, (b) a current state of the one or more manifests determined based on accessibility, regular update, and suitable encoding, (c) digital rights management, (d) compatibility with the first client device, (e) rules provided by a stream owner operator, and (f) user preferences defined in a repository of schedules, rights, and user preferences database.

19. The system according to claim 18, wherein the one or more processors are further configured to:

retrieve alternate metadata for the alternate additional content from a content packaging and distribution system, wherein the alternate metadata indicates a location for the transition from the first additional content to the alternate additional content;

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determine placement of the alternate additional content within the one or more manifests based on the indexed metadata and the alternate metadata;

generate an alternate programming schedule for the first client device based on the alternate additional content and the alternate metadata; and

transmit the alternate programming schedule to another processor.

20. The system according to claim 1, wherein the one or more processors are further configured to:

receive, from a first client device, a request comprising one or more parameters;

determine an alternate additional content for inclusion in the one or more manifests or replace a first additional content in the one or more manifests, wherein the alternate additional content is determined or the first additional content is replaced based on the one or more parameters;

modify the first programming schedule to generate an alternate programming schedule for the first client device, wherein the first programming schedule is modified based on the alternate additional content and associated metadata; and

transmit the alternate programming schedule to at least another processor.

21. The system according to claim 1, wherein the one or more processors are further configured to:

revoke the one or more manifests published for a first client device based on an identifier primitive associated with a media stream of the first client device, wherein the identifier primitive is determined as a plagiarized media player.

22. The system according to claim 1, wherein the one or more manifests are generated for a first client device based on a pre-defined conversion mode, wherein the pre-defined conversion mode corresponds to at least one of a pre-encoded media assets to live stream mode, a pre-encoded media assets to live stream mode with scalable architecture, a live stream to live stream mode, or a mixed mode corresponding to a switch between pre-encoded media assets and live streams.

23. A system, comprising at least one memory for storing instructions and one or more processors configured to:

retrieve indexed metadata associated with at least one of one or more live input streams and one or more pre-encoded media assets referenced by at least one of universal resource locators or identifiers;

generate a first programming schedule comprising one or more decision points defined by one or more content placement opportunities in the indexed metadata; and

control publishing of one or more manifests based on a schedule of the one or more pre-encoded media assets and the one or more decision points in the first programming schedule, wherein the one or more decision points indicate real-time facilitation relative to a playback of a media stream that is streamed on a media player of a first client device, and is associated with the one or more manifests reaching the one or more decision points, of one or more user interactions relating to one or more of content exclusion or content selection;

receive a first manifest request from the first client device, wherein the first manifest request is associated with at least one of the one or more manifests, and comprises one or more parameters, and wherein the one or more parameters comprise an indicator of one or more pre-encoded media assets;

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wherein in response to receiving a first manifest request from a first client device and associated with at least one of the one or more manifests, a pre-encoded media asset from the one or more pre-encoded media assets, based on the indicator of the pre-encoded media asset, is selected in place of the one or more live input streams in an absence of the one or more live input streams; generate an updated manifest comprising the indicator of the pre-encoded media asset; and

publish the updated manifest to a content delivery system for streaming the pre-encoded media asset in place of the one or more live input streams for streaming the pre-encoded media asset via the media player of the first client device.

24. The system according to claim 23, wherein the first client device comprises one or more parameters, and wherein the one or more processors are further configured to:

transmit the first manifest request to another processor; and

identify a customized first programming content and a targeted first non-programming content as a first additional content that is scheduled in the one or more content placement opportunities, wherein the customized first programming content and the targeted first non-programming content are identified based on the one or more parameters in the first manifest request and the indexed metadata, wherein the identifiers refer to records for at least one of the one or more live input streams or the one or more pre-encoded media assets.

25. A computer-implemented method, comprising: generating, by one or more processors, a first programming schedule comprising one or more decision points defined by one or more content placement opportunities in an indexed metadata;

controlling, by the one or more processors, publishing of one or more manifests based on a schedule of one or more pre-encoded media assets and the one or more decision points in the first programming schedule, wherein the one or more decision points indicate real-time facilitation relative to a playback of a media stream that is streamed on a media player of a first client device, and is associated with the one or more manifests reaching the one or more decision points, of one or more user interactions relating to at least one of content exclusion or content selection;

receiving a first manifest request from a first client device, wherein the first manifest request is associated with at least one of the one or more manifests, and comprises one or more parameters, and wherein the one or more parameters comprise an indicator of one or more pre-encoded media assets; and

responsive to receiving the first manifest request:

transmitting, to a second processor, an identifier representing one or more records for at least one of one or more live input streams and the one or more pre-encoded media assets;

in an absence of the one or more live input streams, selecting a pre-encoded media asset from the one or more pre-encoded media assets, based on the indicator of the pre-encoded media asset, in place of the one or more live input streams;

generate an updated manifest comprising the indicator of the pre-encoded media asset; and

publish the updated manifest to the second processor for streaming the pre-encoded media asset in place of the

one or more live input streams for streaming the pre-encoded media asset via the media player of the first client device.

26. The computer-implemented method according to claim 25, wherein the one or more manifests are selected based on the one or more parameters, and the computer-implemented method further comprises:

determining first additional content comprising a customized first programming content and a targeted first non-programming content for the first client device based at least on the one or more parameters.

27. The computer-implemented method according to claim 26, further comprising:

inserting, the first additional content at the one or more content placement opportunities based on the first programming schedule, wherein the one or more content placement opportunities comprise at least one of programming content indicators, non-programming content indicators, graphical treatment indicators, or interactive content indicators.

28. The computer-implemented method according to claim 27, wherein the one or more content placement opportunities comprise programming content indicators, wherein the programming content indicators comprise a plurality of categories.

29. The computer-implemented method according to claim 26, further comprising:

transmitting a request to a content decisioning service for determining the customized first programming content from a content package and distribution system for matching one or more personalization parameters.

30. The computer-implemented method according to claim 26, further comprising modifying, by the one or more processors, a portion of the first programming schedule based on the one or more user interactions in response to receiving the first manifest request.

31. The computer-implemented method according to claim 26, further comprising:

transmitting at least one of universal resource locators or identifiers that reference the one or more pre-encoded media assets to a second processor.

32. The computer-implemented method according to claim 26, further comprising:

retrieving the indexed metadata from a content packaging and distribution system.

33. The computer-implemented method according to claim 26, further comprising inserting based on the first programming schedule, the customized first programming content and the targeted first non-programming content into the one or more manifests based on the one or more content placement opportunities in the indexed metadata and one or more rules and conditions defined in a repository of schedules, rights, and user preferences database.

34. The computer-implemented method according to claim 26, further comprising selecting a second media stream based on accessibility of a media stream, rules provided by a stream owner or a stream operator and user preferences defined in a repository of schedules, rights, and user preferences database, wherein the media stream is unchanged or the media stream has at least one of media or a manifest that is incompatible with a client request.

35. The computer-implemented method according to claim 26, further comprising transitioning from the first additional content to an alternate additional content based on one or more transition parameters.

36. The computer-implemented method according to claim 35, further comprising:

retrieving an alternate metadata for the alternate additional content from a content packaging and distribution system, wherein the alternate metadata indicates a location for the transition from the first additional content to the alternate additional content;

determining, by the one or more processors, placement of the alternate additional content within the one or more manifests based on the indexed metadata and the alternate metadata; and

generating, by the one or more processors, an alternate programming schedule for the first client device based on the alternate additional content and the alternate metadata.

37. The computer-implemented method according to claim 26, further comprising:

receiving a request comprising the one or more parameters from the first client device, determining, by the one or more processors, an alternate additional content for inclusion in the one or more manifests or replacing the first additional content in the one or more manifests, wherein the alternate additional content is determined or the first additional content is replaced based on the one or more parameters;

modifying, by the one or more processors, the first programming schedule to generate an alternate programming schedule for the first client device, wherein the first programming schedule is modified based on the alternate additional content and associated metadata; and

transmitting the alternate programming schedule to a second processor to generate an updated manifest for the first client device.

38. The computer-implemented method according to claim 26, further comprising:

revoking, by the one or more processors, the one or more manifests published for the first client device based on an identifier primitive associated with a media stream of the first client device, wherein the identifier primitive is determined as a plagiarized media player.

39. A non-transitory computer-readable medium including instructions that, when executed by one or more processors, cause the one or more processors to:

generate a first programming schedule comprising one or more decision points defined by one or more content placement opportunities in an indexed metadata;

control publishing of one or more manifests based on a schedule of one or more pre-encoded media assets and the one or more decision points in the first programming schedule, wherein the one or more decision points indicate real-time facilitation relative to a playback of a media stream that is streamed on a media player of a first client device, and is associated with the one or more manifests reaching the one or more decision points, of one or more user interactions relating to one or more of content exclusion or content selection;

receive a first manifest request from a first client device, wherein the first manifest request is associated with at least one of the one or more manifests, and comprises one or more parameters, and wherein the one or more parameters comprise an indicator of one or more pre-encoded media assets; and

responsive to receiving the first manifest request:

transmit, to a second processor, an identifier representing one or more records for at least one of one or more live input streams and the one or more pre-encoded media assets;

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in an absence of the one or more live input streams, select
a pre-encoded media asset from the one or more
pre-encoded media assets, based on the indicator of the
pre-encoded media asset, in place of the one or more
live input streams; and
generate an updated manifest comprising the indicator of
the pre-encoded media asset; and
publish the updated manifest to the second processor for
streaming the pre-encoded media asset in place of the
one or more live input streams for streaming the
pre-encoded media asset via the media player of the
first client device.

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